

10TH MATHS PUBLIC EXAM QUESTIONS COLLECTIONS

10TH MATHS PUBLIC EXAM SEPTEMBER - (2019 -2020)

Time Allowed: 3.00 Hours

Maximum Marks: 100

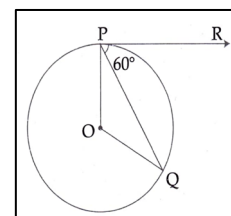
PART - I

Note : (i) Answer all the questions.

14 × 1 = 14

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If $A = \{1,2\}$, $B = \{1,2,3,4\}$, $C = \{5,6\}$ and $D = \{5,6,7,8\}$, then state which of the following statement is true?
 (a) $(A \times C) \subset (B \times D)$ (b) $(B \times D) \subset (A \times C)$
 (c) $(A \times B) \subset (A \times D)$ (d) $(D \times A) \subset (B \times A)$
- Let $f(x) = x^2 - x$, then $f(x-1) - f(x+1)$ is :
 (a) $4x$ (b) $2 - 2x$ (c) $2 - 4x$ (d) $4x - 2$
- Using Euclid's division lemma, if the cube of any positive integer is divided by 9, then the possible remainders are:
 (a) 0,1,8 (b) 1, 4, 8 (c) 0,1,3 (d) 1, 3, 5
- If $A = 2^{65}$ and $B = 2^{64} + 2^{63} + 2^{62} + \dots + 2^0$, which of the following is true?
 (a) B is 2^{64} more than A (b) A and B are equal
 (c) B is larger than A by 1 (d) A is larger than B by 1
- $\frac{a^2}{a^2-b^2} + \frac{b^2}{b^2-a^2} =$
 (a) $a - b$ (b) $a + b$ (c) $a^2 - b^2$ (d) 1
- Transpose of a column matrix is :
 (a) unit matrix (b) diagonal matrix (c) column matrix (d) row matrix
- In $\triangle LMN$, $\angle L = 60^\circ$, $\angle M = 50^\circ$. If $\triangle LMN \sim \triangle PQR$, then the value of $\angle R$ is :
 (a) 40° (b) 70° (c) 30° (d) 110°
- In the figure, if PR is tangent to the circle at P and O is the centre of the circle, then $\angle POQ$ is :
 (a) 120° (b) 100° (c) 110° (d) 90°
- The straight line given by the equation $x = 11$ is
 (a) Parallel to x - axis (b) Parallel to y -axis
 (c) Passing through the origin (d) Passing through the point (0,11)
- If $\tan \theta + \cot \theta = 2$, then the value of $\tan^2 \theta + \cot^2 \theta$ is :
 (a) 0 (b) 1 (c) 2 (d) 4
- A child reshapes a cone made up of clay of height 24 cm and radius 6 cm into a sphere, then the radius of sphere is :
 (a) 24 cm (b) 12 cm (c) 6 cm (d) 48 cm
- A spherical ball of radius r_1 units is melted to make 8 new identical balls each of radius r_2 units. Then $r_1 : r_2$ is :
 (a) 2: 1 (b) 1 : 2 (c) 4 : 1 (d) 1 : 4
- The mean of 100 observations is 40 and their standard deviation is 3. The sum of squares of all deviations is:
 (a) 40000 (b) 160900 (c) 160000 (d) 30000
- If a letter is chosen at random from the English alphabets $\{a, b, c, \dots, z\}$, then the probability that the letters chosen precedes x , is :
 (a) $\frac{12}{13}$ (b) $\frac{1}{13}$ (c) $\frac{23}{26}$ (d) $\frac{3}{26}$



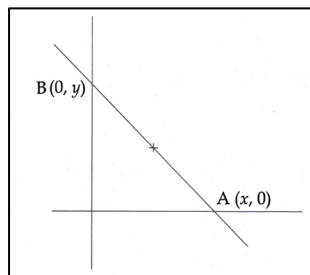
PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

15. If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B.
16. Show that the Function $f: N \rightarrow N$ defined by $f(m) = m^2 + m + 3$ is one-one function.
17. If m, n are natural numbers, for what values of m , does $2^n \times 5^m$ end in 5?
18. Find the 3^{rd} and 4^{th} terms of a sequence, if $a_n = \begin{cases} n^2 & \text{if } n \text{ is odd} \\ \frac{n^2}{2} & \text{if } n \text{ is even} \end{cases}$
19. Find the value of $1^2 + 2^2 + 3^2 + \dots + 10^2$ and hence deduce $2^2 + 4^2 + 6^2 + \dots + 20^2$.
20. Find the value of k for which the equation $9x^2 + 3kx + 4 = 0$ has real and equal roots.
21. If $A = \begin{bmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{bmatrix}$ then find the transpose of $-A$.
22. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 5 \text{ cm}$, $AC = 10 \text{ cm}$
 $BD = 1.5 \text{ cm}$ and $CD = 3.5 \text{ cm}$
23. Find the slope of a line joining the points $(14, 10)$ and $(14, -6)$.
24. Prove that $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} = \sec \theta + \tan \theta$
25. Find the diameter of a sphere whose surface area is 154 m^2 .
26. If the base area of a hemispherical solid is 1386 sq. meters , then find its total surface area?
27. Find the range and coefficient of range of the following data 63, 89, 98, 125, 79, 108, 117, 68
28. Find the volume of the iron used to make a hollow cylinder of height 9 cm and whose internal and external radii are 3 cm and 5 cm respectively.

PART - III**Note : Answer any 10 questions. Question No. 42 is compulsory.****10 × 5 = 50**

29. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8,
 C = The set of even prime number. Verify $(A \cap B) \times C = (A \times C) \cap (B \times C)$
30. Let $A = \{1,2,3,4\}$ and $B = \{2,5,8,11,14\}$ be two sets. Let $f: A \rightarrow B$ be a function give by
 $f(x) = 3x - 1$. Represent this function
(i) by arrow diagram (ii) in a table form
(iii) as a set of ordered pairs (iv) in a graphical form
31. Find the sum of all natural numbers between 100 and 1000 which are divisible by 11.
32. Solve: $6x + 2y - 5z = 13$, $3x + 3y - 2z = 13$, $7x + 5y - 3z = 26$
33. Find the GCD of the given polynomials $x^4 + 3x^3 - x - 3$ and $x^3 + x^2 - 5x + 3$
34. Find the square root of the following polynomials by division method. $\frac{x^2}{y^2} - \frac{10x}{y} + 27 - \frac{10y}{x} + \frac{y^2}{x^2}$
35. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$, and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$
36. State and prove Angle Bisector Theorem.
37. Find the value of ' k ' if the area of quadrilateral is 28 sq. units, whose vertices are $(-4, -2)$,
 $(-3, k)$, $(3, -2)$ and $(2, 3)$.
38. From the top of the tower 60 m high the angles of depression of the top and bottom of a vertical lamp post are observed to be 38° and 60° respectively. Find the height of the lamp post.
($\tan 38^\circ = 0.7813$, $\sqrt{3} = 1.732$).

39. A cylindrical glass with diameter 20 cm has water to a height of 9cm. A small cylindrical metal of radius 5 cm and height 4 cm is immersed it completely. Calculate the rise of the water in the glass.
40. The scores of a cricketer in 7 matches are 70, 80, 60, 60, 40, 90, 95. Find the standard deviation.
41. Two unbiased dice are rolled once. Find the probability of getting
 (i) a doublet (equal numbers on both dice) (ii) the product as a prime number
 (iii) the sum as a prime number (iv) the sum as 1
42. A straight line AB cuts the co-ordinate axes at A and B. If the mid-point of AB is (2, 3) ,find the equation of AB.



PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.

- (a) Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC. (scale factor $\frac{6}{5} > 1$)

OR

- (b) Draw two tangents from a point which is 10 cm away from the center of a circle of radius 5 cm. Also measure the lengths of the tangents.

44.

- (a) Graph the quadratic equation $x^2 - 8x + 16 = 0$ and state the nature of their solution.

OR

- (b) Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$.

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10TH MATHS PUBLIC EXAM SEPTEMBER - (2020-2021)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- If $n(A \times B) = 6$ and $A = \{1,3\}$ then $n(B)$ is :
(a) 1 (b) 2 (c) 3 (d) 6
- The sum of the exponents of the prime factors in the prime factorization of 1729 is
(a) 1 (b) 2 (c) 3 (d) 4
- Given $F_1 = 1, F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is :
(a) 3 (b) 5 (c) 8 (d) 11
- The square root of $\frac{256x^8y^4z^{10}}{25x^6y^6z^6}$ is equal to:
(a) $\frac{16}{5} \left| \frac{x^2z^4}{y^2} \right|$ (b) $16 \left| \frac{y^2}{x^2z^4} \right|$ (c) $\frac{16}{5} \left| \frac{y}{xz^2} \right|$ (d) $\frac{16}{5} \left| \frac{xz^2}{y} \right|$
- Graph of a linear equation is a
(a) Straight line (b) Circle (c) Parabola (d) Hyperbola
- The G.C.D. of a^m, a^{m+1}, a^{m+2} is:
(a) a^m (b) a^{m+1} (c) a^{m+2} (d) 1
- If in $\triangle ABC$, $DE \parallel BC$, $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm then, the length of AE is:
(a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm
- How many tangents can be drawn to the circle from an exterior point?.
(a) one (b) two (c) infinite (d) zero
- The area of a triangle formed by the points $(-5,0)$, $(0,-5)$ and $(5,0)$ is
(a) 0 sq. units (b) 25 sq. units (c) 5 sq. units (d) none of these
- The perimeter of a triangle formed by the points $(0,0)$, $(1,0)$ and $(0,1)$ is:
(a) $\sqrt{2}$ (b) 2 (c) $2 + \sqrt{2}$ (d) $2 - \sqrt{2}$
- If the ratio of the height of a tower and the length of its shadow is $\sqrt{3} : 1$ then, the angle of elevation of the sun has measure :
(a) 45° (b) 30° (c) 90° (d) 60°
- The height of a right circular cone whose radius is 5 cm and slant height is 13 cm will be :
(a) 12 cm (b) 10 cm (c) 13 cm (d) 5 cm
- The total surface area of a hemisphere is how many times the square of its radius?.
(a) π (b) 4π (c) 3π (d) 2π
- A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is :
(a) $\frac{3}{10}$ (b) $\frac{7}{10}$ (c) $\frac{3}{9}$ (d) $\frac{7}{9}$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- If $A = \{1,3,5\}$ and $B = \{2,3\}$ then find. Show that $n(A \times B) = n(B \times A) = n(A) \times n(B)$.
- Let $A = \{1, 2, 3, 4, \dots, 45\}$ and R be the relation defined as “is square of” on A . write R as a subset of $A \times A$. Also, find the domain and range of R .
- Find the number of terms in the A.P. 3, 6, 9, 12, ... 111.
- If $3 + k, 18 - k, 5k + 1$ are in A.P. then find the value of ‘k’.
- Determine the quadratic equation whose sum and product of roots are -9 and 20 .
- Determine the nature of the roots for the quadratic equation $15x^2 + 11x + 2 = 0$.

21. In ΔABC , D and E are points on the sides AB and AC respectively such that $DE \parallel BC$ If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15 \text{ cm}$ find AE .
22. Show that the given points are collinear $(-3, -4)$, $(7, 2)$ and $(12, 5)$.
23. Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$.
24. Find the intercepts made by the following lines on the coordinate axes. $3x - 2y - 6 = 0$.
25. Find the angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of a tower of height $10\sqrt{3} \text{ m}$.
26. Find the volume of a cylinder whose height is 2 m and whose base area is 250 m^2 .
27. A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head.
28. The heights of two right circular cones are in the ratio $1:2$ and the perimeters of their bases are in the ratio $3:4$. Find the ratio of their volumes.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{x \in W | x < 2\}$, $B = \{x \in N | 1 < x \leq 4\}$ and $C = \{3, 5\}$. Verify that
(i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
30. The sum of three consecutive terms that are in A.P. is 27 and their product is 288. Find the three terms.
31. Find the HCF of 396, 504, 636.
32. Solve $x + y + z = 5$, $2x - y + z = 9$, $x - 2y + 3z = 16$.
33. Find the square root of $64x^2 - 16x^3 + 17x^2 - 2x + 1 = 0$.
34. State and prove Pythagoras Theorem.
35. Show that in a triangle, the medians are concurrent.
36. Find the equation of the median and altitude of ΔABC through A where the vertices are $A(6, 2)$, $B(-5, -1)$ and $C(1, 9)$.
37. If the points $P(-1, -4)$, $Q(b, c)$ and $R(5, -1)$ are collinear and if $2b + c = 4$, then find the values of ' b ' and ' c '.
38. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$).
39. If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7 cm , find the volume of the frustum.
40. A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25 cm . Find the total surface area of the toy if its common diameter is 12 cm .
41. Two dice are rolled. Find the probability that sum of outcomes is
(i) equal to 4, (ii) greater than 10, (iii) less than 13.
42. If the equation $(1 + m^2)x^2 + 2mcx + c^2 - a^2 = 0$ has equal roots, then prove that
 $c^2 = a^2(1 + m^2)$

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.

- (a) Construct a ΔPQR whose base $PQ = 4.5 \text{ cm}$, $\angle R = 35^\circ$ and the median from R to PQ is 6 cm .

OR

- (b) Draw a circle of diameter 6 cm from a point P , which is 8 cm away from its centre. Draw two tangents PA and PB to the circle and measure their lengths.

44.

- (a) Draw the graph of $x^2 + x - 12 = 0$ and state the nature of their solution.

OR

- (b) Draw the graph of $y = x^2 + 3x - 4$ and hence use it to solve $x^2 + 3x - 4 = 0$.

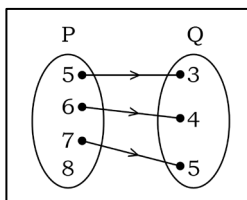


10TH MATHS PUBLIC EXAM MAY - (2021-2022)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

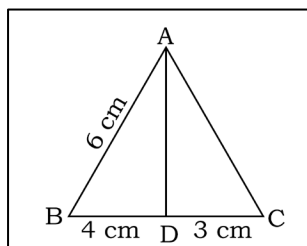
- If the ordered pairs $(a + 2, 4)$ and $(5, 2a + b)$ are equal then (a, b) is :
 (a) $(2, -2)$ (b) $(5, 1)$ (c) $(2, 3)$ (d) $(3, -2)$
- If the HCF of 65 and 117 is expressible in the form of $65m - 117$, then the value of 'm' is :
 (a) 4 (b) 2 (c) 1 (d) 3
- If t_n is the n^{th} term of a A.P., then $t_{8n} - t_n$ is :
 (a) $(8n - 1)d$ (b) $(8n - 2)d$ (c) $(7n - 2)d$ (d) $(7nd)$
- If $(x - 6)$ is the HCF of $\beta - 2x - 24$ and $\beta - kx - 6$, then the value of k is
 (a) 3 (b) 5 (c) 6 (d) 8
- Which of the following should be added to make $x^4 + 64$ a perfect square?
 (a) $4x^2$ (b) $16x^2$ (c) $8x^2$ (d) $-8x^2$
- The number of points of intersection of the quadratic polynomial $x^2 + 4x + 4$ with the X-axis is:
 (a) 0 (b) 1 (c) 0 or 1 (d) 2
- If $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is :
 (a) 2.5 cm (b) 5 cm (c) 10 cm (d) $5\sqrt{2}$ cm
- In a $\triangle ABC$, AD is the bisector of $\angle BAC$. If $AB = 8$ cm, $BD = 6$ cm and $DC = 3$ cm, length of the side AC is :
 (a) 6 cm (b) 4 cm (c) 3 cm (d) 8 cm
- If $(5, 7)$, $(3, p)$ and $(6, 6)$ are collinear, then the value of 'p' is :
 (a) 3 (b) 6 (c) 9 (d) 12
- The slope of the line which is perpendicular to a line joining the points $(0, 0)$ and $(-8, 8)$ is:
 (a) -1 (b) 1 (c) $\frac{1}{3}$ (d) -8
- A tower is 60 m high. Its shadow is x metres shorter when the sun's altitude is 45° than when it had been 30° , then 'x' is equal to :
 (a) 41.92 m (b) 43.92 m (c) 43m (d) 45.6 m
- If two solid hemispheres of same base radius 'r' units are joined together along their bases, then curved surface area of this new solid is :
 (a) $4\pi r^2$ sq. units (b) $6\pi r^2$ sq. units (c) $3\pi r^2$ sq. units (d) $8\pi r^2$ sq. units
- If the radius of the cylinder is doubled, the new volume of the cylinder will be times the original volume.
 (a) same (b) 3 (c) 4 (d) 2
- The probability of getting a job for a person is $\frac{x}{3}$. If the probability of not getting the job is $\frac{2}{3}$, then the value of x is :
 (a) 2 (b) 1 (c) 3 (d) 1.5

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- Let $A = \{1, 2, 3\}$ and $B = \{x | x \text{ is a prime number less than } 10\}$. Find $A \times B$ and $B \times A$.
- The arrow diagram shows in figure a relationship between the sets P and Q . Write the relation in
 (i) Set builder form (ii) Roster form (iii) What is the domain and range of R.



17. If $13824 = 2^a \times 3^b$ then find 'a' and 'b'.
18. Which term of an A.P. 16, 11, 6, 1,... is -54 ?
19. Find the excluded values of the following expression $\frac{7p+2}{8p^2+13p+5}$.
20. In the figure AD is the bisector of $\angle A$. If $BD = 4\text{ cm}$, $DC = 3\text{ cm}$ and $AB = 6\text{ cm}$, find AC



21. Show that the points $P(-1.5, 3)$, $Q(6, -2)$, $R(-3, 4)$ are collinear.
22. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ?
23. Find the equation of a straight line which has slope $\frac{-5}{4}$ and passing through the point $(-1, 2)$.
24. From the top of a rock $50\sqrt{3}\text{ m}$ height, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock.
25. The radius of a spherical balloon increases from 12 cm to 16 cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases.
26. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights.
27. Two coins are tossed together. What is the probability of getting different faces of the coins?
28. If $P = \frac{x}{x+y}$, $Q = \frac{y}{x+y}$ then find $\frac{1}{(P^2 - Q^2)}$.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime number. Verify $A \times (B - C) = (A \times B) - (A \times C)$.
30. If l^{th} , m^{th} and n^{th} terms of an A.P. are, x , y , z respectively, then show that
 (i) $x(m - n) + y(n - l) + z(l - m) = 0$ (ii) $(x - y)n + (y - z)l + (z - x)m = 0$
31. The ratio of 6^{th} and 8^{th} term of an A.P. is 7: 9. Find the ratio 9^{th} term to 13^{th} term.
32. If $36x^4 - 60x^3 + 61x^2 - mx + n$ is a perfect square, find the values of 'm' and 'n'.
33. Solve $pqx^2 - (p + q)^2x + (p + q)^2 = 0$.
34. If α and β are the roots of $7x^2 + ax + 2 = 0$ and $\beta - \alpha = \frac{-13}{7}$. Find the values of a .
35. State and Prove Basic Proportionality Theorem (BPT) or Thales theorem.
36. An Aeroplane leaves an airport and flies due north at a speed of 1000 km/hr . At the same time, another Aeroplane leaves the same airport and flies due west at the speed of 1200 km/hr . How far apart will be the two planes after $1\frac{1}{2}$ hours?.
37. A quadrilateral has vertices $A(-4, -2)$, $B(5, -1)$, $C(6, 5)$ and $D(-7, 6)$. Show that the mid-points of its sides form a parallelogram.

38. From a point on the ground, the angles of elevation of the bottom and top of a tower fixed at the top of a 30 m high building are 45° and 60° respectively. Find the height of the tower.
($\sqrt{3} = 1.732$).
39. A container open at the top is in the form of a frustum of a cone of height 16 cm with radii of its lower and upper ends are 8 cm and 20 cm respectively. Find the cost of milk which can completely fill a container at the rate of 540 per liter.
40. Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3 cm and its length 12 cm. If each cone has height 2 cm, find the volume of the model that Nathan made.
41. In a class of 50 students, 28 opted for NCC, 30 opted for NSS and 18 opted both NCC and NSS. One of the students is selected at random. Find the probability that
(i) The student opted for NCC but not NSS. (ii) The student opted for NSS but not NCC.
(iii) The student opted for exactly one of them.
42. Find the equation of the line passing through (22, -6) and having intercept on X- axis exceeds the intercept on Y-axis by 5 units.

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.
(a) Construct a $\triangle ABC$ such that $AB = 5.5$ cm, $\angle C = 25^\circ$ and the altitude from C to AB is 4 cm.
OR
(b) Draw the two tangents from a point which is 5 cm away from the centre of a circle of diameter 6 cm. Also, measure the lengths of the tangents.
44.
(a) Draw the graph of $y = x^2 - 4x + 3$ and use it to solve $x^2 - 6x + 9 = 0$.
OR
(b) Draw the graph of $x^2 - 4x + 4 = 0$ and state the nature of their solution.



10TH MATHS PUBLIC EXAM AUGUST - (2021-2022)

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

 $14 \times 1 = 14$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If there are 1024 relations from a set $A = \{1, 2, 3, 4, 5\}$ to set B , then the number of elements in B is:
(a) 3 (b) 2 (c) 4 (d) 8
- The range of the Relation $R = \{ \{x \mid x^2\}, x \text{ is a Prime number less than } 13 \}$
(a) $\{2, 3, 5, 7\}$ (b) $\{2, 3, 5, 7, 11\}$ (c) $\{4, 9, 25, 49, 121\}$ (d) $\{1, 4, 9, 25, 49, 121\}$
- The sum of the exponents of the prime factors in the prime factorization 1729 is:
(a) 1 (b) 2 (c) 3 (d) 4
- A system of three linear equations in the three variables is inconsistent if their planes:
(a) Intersect only at a point (b) Intersect in a line
(c) Coincide with each other (d) Do not intersect
- The solution of the system $x + y - 3z = -6, -7y + 7z = 7, 3z = 9$ is
(a) $x = 1, y = 2, z = 3$ (b) $x = -1, y = 2, z = 3$
(c) $x = -1, y = -2, z = 3$ (d) $x = 1, y = -2, z = 3$
- $y^2 + \frac{1}{y^2}$ is not equal to ::
(a) $\frac{y^4+1}{y^2}$ (b) $\left(y + \frac{1}{y}\right)^2$ (c) $\left(y - \frac{1}{y}\right)^2 + 2$ (d) $\left(y + \frac{1}{y}\right)^2 - 2$
- If in $\triangle ABC$, $DE \parallel BC$, $AB = 3.6\text{cm}$, $AC = 2.4\text{cm}$ and $AD = 2.1\text{cm}$ then, the length of AE is :
(a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm
- How many tangents can be drawn to the circle from an Exterior Point
(a) one (b) two (c) infinite (d) zero
- The point of intersection of $3x - y = 4$ and $x + y = 8$ is :
(a) (5, 3) (b) (2, 4) (c) (3, 5) (d) (4, 4)
- If slope of the line PQ is $\frac{1}{\sqrt{3}}$ then, slope of the perpendicular bisector of PQ is
(a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) 0
- The angle of elevation of a cloud from a point h metre above a lake is S . The angle of depression of its reflection in the lake is 45° . The height of location of the cloud from the lake (in meters) is :
(a) $\frac{h(1 + \tan \beta)}{1 - \tan \beta}$ (b) $\frac{h(1 - \tan \beta)}{1 + \tan \beta}$ (c) $h \tan (45^\circ - \beta)$ (d) None of these
- If the radius of the base of a right circular cylinder is halved keeping the same Height then, the ratio of the volume of the cylinder thus obtained to the volume of original cylinder is :
(a) 1 : 2 (b) 1 : 4 (c) 1 : 6 (d) 1 : 8

13. The total surface area of hemi-sphere is how much times the square of its radius :
 (a) π (b) 4π (c) 3π (d) 2π
14. A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is :
 (a) $\frac{3}{10}$ (b) $\frac{7}{10}$ (c) $\frac{3}{9}$ (d) $\frac{7}{9}$

PART - II

Note : Answer any 10 questions. Question No. 28 is compulsory.

10 × 2 = 20

15. If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B.
16. If $A = \{5,6\}$, $B = \{4,5,6\}$, $C = \{5,6,7\}$. Show that $A \times A = (B \times B) \cap (C \times C)$.
17. Find the least number that is divisible by the first ten natural numbers.
18. Find the 19th term of an A.P. $-11, -15, -19, \dots$
19. Find the square root of the following rational expressions. $\frac{400x^4y^{12}z^{16}}{100x^8y^4z^4}$.
20. ABCD is a trapezium in which $AB \parallel DC$ and P, Q are points on AD and BC respectively, such that $PQ \parallel DC$ if $PD = 18\text{cm}$, $BQ = 35\text{cm}$ and $QC = 15\text{cm}$, find AD.
21. If area of triangle formed by vertices $A(-1,2)$, $B(k, -2)$ and $C(7,4)$ is 22 sq. units. find the value of 'k'.
22. The line p passes through the points $(3, -2)$, $(12,4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q?.
23. Find the slope of a line joining the given points $(5, \sqrt{5})$ with the origin.
24. Find the angle of elevation of the top of a tower from a point on the ground, which is 30m away from the foot of a tower of height $10\sqrt{3}\text{m}$.
25. Find its radius and height. radius 7 cm is 704 cm^2 , then find its slant height.
26. If the total surface area of a cone of The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500 sq. cm.
27. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is (i) white ball (ii) black or red ball.
28. Find the value of 'x', in $x^2 - 4x - 12 = 0$.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Represent each of the given relations by (a) an arrow diagram, (b) a graph and (c) a set in roster form, wherever possible. $\{(x, y) | y = x + 3, x, y \text{ are natural numbers} < 10\}$.
30. Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case.

31. In an A.P nine times ninth term is equal to the fifteen times fifteenth term, show that six times twenty fourth term is zero.
32. Simplify: $\frac{b^2+3b-28}{b^2+4b+4} \div \frac{b^2-49}{b^2-5b-14}$.
33. Find the square root of $x^4 - 12x^3 + 42x^2 - 36x + 9$.
34. Solve $x^2 + 2x - 2 = 0$ by Formula method.
35. State and prove Angle Bisector Theorem.
36. A man goes 18 m due east and then 24 m due north. Find the distance of his current position from the starting point?.
37. Find the area of the quadrilateral formed by the points (8, 6) , (5, 11) , (−5,12) and (−4,3).
38. To a man standing outside his house, the angles of elevation of the top and bottom of a window are 60° and 45° respectively. If the height of the man is 180 cm and if he is 5m away from the wall, what is the height of the window ?. ($\sqrt{3} = 1.732$).
39. A cylindrical drum has height of 20cm and base radius of 14cm. Find its curved surface area and the total surface area.
40. If the circumference of base of a conical wooden piece is 484 cm then find its volume when its height is 105 cm.
41. Two unbiased dice are rolled once. Find the probability of getting
 (i) a doublet (equal numbers on both dice) (ii) the product as a prime number
 (iii) the sum as a prime number (iv) the sum as 1
42. A cat is located at the point (−6, −4) in xy plane. A bottle of milk is kept at (5, 11) . The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.

PART - IV**Note :** Answer the following questions..**2 × 8 = 16**

43.

- (a) Construct a triangle similar to a triangle PQR with its sides equal to $\frac{7}{3}$ of the corresponding sides of the triangle PQR. (Scale factor $\frac{7}{3} > 1$)

OR

- (b) Draw a circle of diameter 6 cm. From a point P , which is 8 cm away from its centre, draw the two tangents PA and PB to the circle and measure their lengths.

44.

- (a) Draw the graph of $x^2 - 9x + 20 = 0$ and state the nature of their solution.

OR

- (b) Draw the graph of $y = x^2 - 4x + 3$ and use it to solve $x^2 - 6x + 9 = 0$.



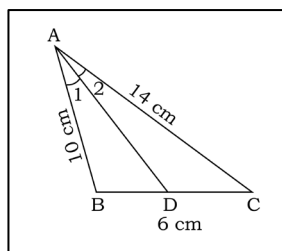
10TH MATHS PUBLIC EXAM APRIL - (2022-2023)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- $A = \{a, b, p\}$, $B = \{2, 3\}$, $C = \{p, q, r, s\}$ then $n[(A \cup C) \times B]$ is:
(a) 8 (b) 20 (c) 12 (d) 16
- If $n(A) = p$, $n(B) = q$. then the total number of relations that exist from A to B is
(a) 0 (b) 1 (c) $2^{pq} - 1$ (d) 2^{pq}
- Given $F_1 = 1$, $F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is :
(a) 3 (b) 5 (c) 8 (d) 11
- If the sequence t_1, t_2, t_3 are in A.P., then be sequence t_6, t_{12}, t_{18} ...
(a) a Geometric progression (b) an Arithmetic Progression
(c) neither an Arithmetic Progression nor a Geometric progression
(d) a constant sequence
- $\frac{3y-3}{y} \div \frac{7y-7}{3y^2}$ is :
(a) $\frac{9y}{7}$ (b) $\frac{9y^3}{(21y-21)}$ (c) $\frac{21y^2-42y+21}{3y^3}$ (d) $\frac{7(y^2-2y+1)}{y^2}$
- Graph of a Quadratic Equation is a
(a) straight line (b) circle (c) parabola (d) hyperbola
- If in triangles ABC and EDF, $\frac{AB}{DE} = \frac{BC}{FD}$ then they will be similar when :
(a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$
- A tangent of a circle is perpendicular to the radius at the
(a) centre (b) point of contact (c) infinity (d) Chord
- The slope of the straight line perpendicular to x-axis is :
(a) 1 (b) 0 (c) ∞ (d) -1
- If $\sin \theta = \cos \theta$ then the value of $2 \tan^2 \theta + \sin^2 \theta - 1$ is :
(a) $\frac{3}{2}$ (b) $\frac{-3}{2}$ (c) $\frac{2}{3}$ (d) $\frac{-2}{3}$
- The height of a right circular cone whose radius is 5 cm and slant height is 13 cm will be :
(a) 12 cm (b) 10 cm (c) 13 cm (d) 5 cm
- The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is :
(a) 1:2:3 (b) 2:1:3 (c) 1:3:2 (d) 3:1:2
- If the sum and mean of a data are 407 and 11 respectively, then the number of observations in the data are:
(a) 37 (b) 4477 (c) 396 (d) 418
- If a letter is chosen at random from the English alphabets $\{a, b, c, \dots, z\}$ then the probability that the letter chosen precedes x:
(a) $\frac{12}{13}$ (b) $\frac{1}{13}$ (c) $\frac{23}{26}$ (d) $\frac{3}{26}$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- If $B \times A = \{(-2, 3), (-2, 4), (0, 3), (0, 4), (3, 3), (3, 4)\}$ find A and B.
- Find 'k' if $f \circ f(k) = 5$ where $f(k) = 2k - 1$.

17. Find 'x' so that $x + 6$, $x + 12$ and $x + 15$ are consecutive terms of a Geometric Progression.
18. Simplify: $\frac{x+2}{4y} \div \frac{x^2-x-6}{12y^2}$.
19. Determine the nature of roots for the following quadratic equation. $2x^2 - x - 1 = 0$.
20. In the figure AD is the bisector of $\angle BAC$, if $AB = 10 \text{ cm}$, $AC = 14 \text{ cm}$ and $BC = 6 \text{ cm}$. Find BD and DC.



21. A cat is located at the point $(-6, -4)$ in xy plane. A bottle of milk is kept at $(5, 11)$. The cat wishes to consume the milk travelling through shortest possible distance. Find the equation of the path it needs to take its milk.
22. If the straight lines $12y = -(p+3)x + 12$, $12x - 7y = 16$ are perpendicular then find 'p'.
23. Prove that $\frac{\sec \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = \cot \theta$.
24. The radius of a conical tent is 7 m and height is 24 m . Calculate the length of the canvas used to make the tent if the width of the rectangular canvas is 4 m ?
25. If the ratio of radii of two spheres is $4:7$, find the ratio of their volumes.
26. Find the range and coefficient of range of the following data 63, 89, 98, 125, 79, 108, 117, 68.
27. A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3. Prove that the probability of B being selected is atmost 0.8.
28. If $p^2 \times q^1 \times r^4 \times s^3 = 3,15,000$ then find the values of 'p', 'q', 'r' and 's'

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2, 4, 6, 10, 12\}$, $B = \{0, 1, 2, 4, 5, 9\}$. Represent f by
 (i) Set of ordered pairs, (ii) a table (iii) a graph (iv) an arrow diagram
30. The houses of a street are numbered from 1 to 49. Senthil's house is numbered Such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number?.
31. Find the sum to n terms of the series $5 + 55 + 555 + \dots$
32. Solve: $x + 20 = \frac{3y}{2} + 10 = 2z + 5 = 110 - (y + z)$.
33. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$.
34. Two poles of height 'a' meters and 'b' meters are 'p' meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters.
35. State and prove Angle Bisector Theorem.
36. Find the area of the quadrilateral formed by the points $(8, 6)$, $(5, 11)$, $(-5, 12)$ and $(-4, 3)$.
37. Find the equation of a straight line parallel to X - axis and passing through the point of intersection of the lines $7x - 3y = -12$ and $2y = x + 3$.

38. From the top of a lighthouse, the angle of depression of two ships on the opposite sides of it are observed to be 30° and 60° . If the height of the lighthouse is h meters and the line joining the ships passes through the foot of the lighthouse, show that the distance between the ships is $\frac{4h}{\sqrt{3}}$ m.
39. The radius and height of a cylinder are in the ratio 5: 7 and its curved surface area is 5500 sq. cm. Find its radius and height.
40. Arul has to make arrangements for the accommodation of 150 persons for his family function. For this purpose, he plans to build a tent which is in the shape of cylinder surmounted by cone. Each person occupies 4 sq. m of the space on ground and 40 cu. Meter of air to breathe. What should be the height of the conical part of the tent if the height of cylindrical part is 8 m?
41. Two unbiased dice are rolled once. Find the probability of getting
 (i) a doublet (equal numbers on both dice) (ii) the product as a prime number
 (iii) the sum as a prime number (iv) the sum as 1
42. Let $A = \{x \in W \mid x < 3\}$, $B = \{x \in N \mid 1 < x \leq 5\}$ and $C = \{3, 5, 7\}$.
 Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.
 (a) Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw two tangents to the circle from that point.
OR
 (b) Draw a triangle ABC of base $BC = 8$ cm. $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $BD = 6$ cm.
44.
 (a) Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the diameter and circumference (approximately related) of each circle as shown in the table and use it to find the circumference of a circle when its diameter is 6 cm.

Diameter (x) cm	1	2	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

OR

- (b) Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$.

♣♣♣♣

10TH MATHS PUBLIC EXAM JULY - (2022-2023)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- If there are 1024 relations from a set $A = \{1, 2, 3, 4, 5\}$ to a set B , then the number of elements in B is:
 (a) 3 (b) 2 (c) 4 (d) 8
- $7^{4k} \equiv \underline{\hspace{1cm}} \pmod{100}$
 (a) 1 (b) 2 (c) 3 (d) 4
- The next term of the sequence $\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \frac{1}{14}, \dots$ is :
 (a) $\frac{1}{15}$ (b) $\frac{1}{16}$ (c) $\frac{1}{18}$ (d) $\frac{1}{20}$
- $y^2 + \frac{1}{y^2}$ is not equal to ::
 (a) $\frac{y^4+1}{y^2}$ (b) $\left(y + \frac{1}{y}\right)^2$ (c) $\left(y - \frac{1}{y}\right)^2 + 2$ (d) $\left(y + \frac{1}{y}\right)^2 - 2$
- Graph of a Linear Equation is a
 (a) straight line (b) circle (c) parabola (d) hyperbola
- If in $\triangle ABC$, $DE \parallel BC$, $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm then, the length of AE is :
 (a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm
- How many tangents can be drawn to the circle from an exterior point ?
 (a) one (b) two (c) infinite (d) zero
- The straight line given by the equation $x = 11$ is :
 (a) parallel to X - axis (b) parallel to y -axis
 (c) passing through the origin (d) passing through the point (0,11)
- If the slope of the line PQ is $\frac{1}{\sqrt{3}}$, then the slope of the perpendicular bisector of PQ is:
 (a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) 0
- $\tan \theta \operatorname{cosec}^2 \theta - \tan \theta$ is equal to
 (a) $\sec \theta$ (b) $\cot^2 \theta$ (c) $\sin \theta$ (d) $\cot \theta$
- The total surface area of hemi-sphere is how much times the square of its radius :
 (a) π (b) 4π (c) 3π (d) 2π
- The curved surface area of the right circular cone of height 15 cm and base diameter is 16 cm:
 (a) $60 \pi \text{ cm}^2$ (b) $68 \pi \text{ cm}^2$ (c) $120 \pi \text{ cm}^2$ (d) $136 \pi \text{ cm}^2$
- The range of the data 8, 8, 8, ..., 8 is
 (a) 0 (b) 1 (c) 8 (d) 3
- The probability of a red marble selected at random from a jar containing p red, q blue and r green marbles is:
 (a) $\frac{q}{p+q+r}$ (b) $\frac{p}{p+q+r}$ (c) $\frac{p+q}{p+q+r}$ (d) $\frac{p+r}{p+q+r}$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- A Relation R is given by the set $\{(x, y)/y = x + 3, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and the range.

16. Check whether $(f \circ g) = (g \circ f)$ if $f(x) = x - 6$ and $g(x) = x^2$.
17. Find the least number that is divisible by the first ten natural numbers.
18. Find the 8th term of the G.P. 9, 3, 1, ...
19. Determine the nature of the roots for the quadratic equation $15x^2 + 11x + 2 = 0$.
20. If $A = \begin{pmatrix} 5 & 2 & 2 \\ -\sqrt{17} & 0.7 & \frac{5}{2} \\ 8 & 3 & 1 \end{pmatrix}$ then verify $(A^T)^T = A$.
21. Check whether AD is bisector of $\angle A$ of $\triangle ABC$ in each of the following $AB = 4 \text{ cm}$, $AC = 6 \text{ cm}$ $BD = 1.6 \text{ cm}$ and $CD = 2.4 \text{ cm}$.
22. Find the slope of a line joining the given points $(5, \sqrt{5})$ with the origin.
23. Prove that $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$.
24. The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2 Find the diameter of the cylinder.
25. The volume of a solid right circular cone is 11088 cm^3 . If its height is 24 cm then find the radius of the cone.
26. Find the standard deviation of first 21 natural numbers.
27. A die is rolled and a coin is tossed simultaneously. Find the probability that the die shows an odd number and the coin shows a head.
28. Find the equation of a straight line which is parallel to the line $3x - 7y = 12$ and passing through the point $(6, 4)$.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{x \in W | x < 2\}$, $B = \{x \in N | 1 < x \leq 4\}$ and $C = \{3, 5\}$.
Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$
30. Find the sum to n terms of the series $3 + 33 + 333 + \dots$
31. Rekha has 15 square Colour papers of sizes 10 cm , 11 cm , 12 cm , ... 24 cm . How much area can be decorated with these Colour papers.
32. Solve $3x - 2y + z = 2$, $2x + 3y - z = 5$, $x + y + z = 6$.
33. Find the square root of the following polynomials by division method.
$$121x^4 - 198x^3 - 183x^2 + 216x + 144.$$
34. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$.
35. State and prove Pythagoras Theorem.
36. Find the area of the quadrilateral whose vertices are at $(-9, -2)$, $(-8, -4)$, $(2, 2)$ and $(1, -3)$
37. Find the equation of the perpendicular bisector of the line joining the points $A(-4, 2)$ and $B(6, -4)$.

38. Prove that $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} + \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = 2 \sec \theta$.

39. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm.

How many small spheres can be obtained ?.

40. Find the coefficient of variation 24, 26, 33, 37, 29, 31.

41. Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8.

42. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$).

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.

(a) Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR . [Scale factor $\frac{3}{5} < 1$]

OR

(b) Draw two tangents from a point which is 10 cm away from the centre of a Circle of radius 5 cm. Also, measure the lengths of the tangents.

44.

(a) Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find y when $x = 9$ (ii) find x when $y = 7.5$.

OR

(b) Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$.



10TH MATHS PUBLIC EXAM APRIL - (2023-2024)

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

 $14 \times 1 = 14$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If $n(A \times B) = 6$ and $A = \{1,3\}$, then $n(B)$ is:
(a) 1 (b) 2 (c) 3 (d) 6
- If $f: A \rightarrow B$ is a bijective function and if $n(B) = 7$ then $n(A)$ is equal to.
(a) 7 (b) 49 (c) 1 (d) 14
- The least number that is divisible by all the numbers from 1 to 10 (inclusive) is:
(a) 2025 (b) 5220 (c) 5025 (d) 2520
- An A.P, consists of 31 terms. If its 16th term is m , then the sum of all the terms of this A.P. is
(a) $16m$ (b) $62m$ (c) $31m$ (d) $\frac{31}{2}m$
- Which of the following should be added to make $x^4 + 64$ a perfect square ?
(a) $4x^2$ (b) $16x^2$ (c) $8x^2$ (d) $-8x^2$
- Graph of a Linear Equation is a
(a) straight line (b) circle (c) parabola (d) hyperbola
- If in $\triangle ABC$, $DE \parallel BC$, $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm then, the length of AE is :
(a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm
- How many tangents can be drawn to the circle from an exterior point?
(a) One (b) Two (c) Infinite (d) Zero
- The area of triangle formed by the points $(-5,0)$, $(0,-5)$ and $(5,0)$ is
(a) 0 sq. units (b) 25 sq. units (c) 5 sq. units (d) 10 sq. units
- If $x = a \tan \theta$ and $y = b \sec \theta$, then:
(a) $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ (b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (d) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$
- The curved surface area of a right circular cylinder of height 4 cm and base diameter 10 cm is
(a) 40π sq. units (b) 20π sq. units (c) 14π sq. units (d) 80π sq. units
- The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is :
(a) 1:2:3 (b) 2:1:3 (c) 1:3:2 (d) 3:1:2
- Which of the following values cannot be a probability of an event?
(a) 0 (b) 0.5 (c) 1.05 (d) 1
- The probability of getting a job for a person is $\frac{x}{3}$. If the probability of not getting the job is $\frac{2}{3}$, then the value of x is :
(a) 2 (b) 1 (c) 3 (d) 1.5

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

15. If $A \times B = \{(3,2), (3,4), (5,2), (5,4)\}$ then find A and B.
16. If $f(x) = 3x - 2$, $g(x) = 2x + k$ and $f \circ g = g \circ f$, then find value of k .
17. 'a' and 'b' are two positive integers such that $a^b \times b^a = 800$. Find 'a' and 'b'
18. Simplify- $\frac{4x^2y}{2z^2} \times \frac{6xz^3}{20y^4}$.
19. Find the sum and product of the roots for following quadratic equation $x^2 + 8x - 65 = 0$.
20. A man goes to 18 m due east and then 24 m due North. Find the distance of his current position from the starting point.
21. If the points $A(-3,9)$, $B(a,b)$ and $C(4,-5)$ are collinear and if $a + b = 1$, then find a and b .
22. Find the equation of a straight line which has slope $-\frac{5}{4}$ and passing through the points $(-1,2)$.
23. Prove that $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$.
24. If the base area of a hemispherical solid is 1386 sq. meters, then find its total surface area?
25. Find the volume of a cylinder whose height is 2m and whose base area is 250 sq.m.
26. Find the Range and Co-efficient of range of the following data : 25,67,48,53,18,39,44.
27. What is the Probability that a leap year selected at random will contain 53 Saturdays?.
28. Find the HCF of 23 and 12.

PART - III**Note : Answer any 10 questions. Question No. 42 is compulsory.****10 × 5 = 50**

29. Let $A = \{x \in N | 1 < x < 4\}$, $B = \{x \in W | 0 \leq x < 2\}$ and $C = \{x \in W | x < 3\}$.
Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
30. Let $A = \{0,1,2,3\}$ and $B = \{1,3,5,7,9\}$ be two sets. Let $f: A \rightarrow B$ be a function give by $f(x) = 2x - 1$. Represent this function
(i) by arrow diagram (ii) in a table form
(iii) as a set of ordered pairs (iv) in a graphical form
31. Find the Sum of $9^3 + 10^3 + \dots + 21^3$.
32. Find the square root of $64x^2 - 16x^3 + 17x^2 - 2x + 1 = 0$.
33. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$.
34. State and Prove Thales Theorem.
35. Find the area of the quadrilateral whose vertices are at $(-9, -2)$, $(-8, -4)$, $(2,2)$ and $(1, -3)$.
36. Find the equation of the perpendicular bisector of the line joining the points $A(-4,2)$ and $B(6, -4)$.

37. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. ($\sqrt{3} = 1.732$).
38. If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7 cm, find the volume of the frustum.
39. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice-cream. The ice-cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container.
40. Find the coefficient of variation 24, 26, 33, 37, 29, 31.
41. Two dice are rolled once. Find the probability of getting an even number on the first die or a total of face sum 8.
42. Find the sum to n terms of the series $7 + 77 + 777 + \dots$

PART - IV**Note : Answer the following questions..** **$2 \times 8 = 16$** **43.**

- (a) Construct a ΔPQR which has base $PQ = 4.5$ cm, $\angle R = 35^\circ$ and the median RG from R to PQ is 6 cm

OR

- (b) Draw a circle of diameter 6 cm, from a point P , which is 8 cm away from its centre. Draw a two tangents PA and PB to the circle and measure their lengths.

44.

- (a) Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$.

OR

- (b) Draw the graph of $xy = 24$, $x, y > 0$. Using the graph find
- (i) y when $x = 3$ and
- (ii) x when $y = 6$.



10TH MATHS PUBLIC EXAM JULY - (2023-2024)

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

 $14 \times 1 = 14$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$, then $f \circ g$ is :
 (a) $\frac{3}{2x^2}$ (b) $\frac{2}{3x^2}$ (c) $\frac{2}{9x^2}$ (d) $\frac{1}{6x^2}$
- If $\{(a, 8), (6, b)\}$ represents an identity function, then the value of a and b are respectively
 (a) (8, 6) (b) (8, 8) (c) (6, 8) (d) (6, 6)
- $7^{4k} \equiv \underline{\hspace{1cm}} \pmod{100}$
 (a) 1 (b) 2 (c) 3 (d) 4
- $\frac{x}{x^2-25} - \frac{8}{x^2+6x+5}$ gives :
 (a) $\frac{x^2-7x+40}{(x-5)(x+5)}$ (b) $\frac{x^2+7x+40}{(x-5)(x+5)(x+1)}$ (c) $\frac{x^2-7x+40}{(x^2-25)(x+1)}$ (d) $\frac{x^2+10}{(x^2-25)(x+1)}$
- Transpose of a column matrix is:
 (a) unit matrix (b) diagonal matrix (c) column matrix (d) row matrix
- Two poles of heights 6 m and 11 m stand vertically on a plane ground. If the distance between their feet is 12 m, what is the distance between their tops?
 (a) 13 m (b) 14 m (c) 15 m (d) 12.8 m
- A tangent is perpendicular to the radius at the:
 (a) centre (b) point of contact (c) infinity (d) Chord
- The straight line given by the equation $x = 11$ is
 (a) Parallel to X - axis (b) Parallel to Y-axis
 (c) Passing through the origin (d) Passing through the point (0, 11)
- The equation of a line passing through the origin and perpendicular to the line $7x - 3y + 4 = 0$ is:
 (a) $7x - 3y + 4 = 0$ (b) $3x - 7y + 4 = 0$ (c) $3x + 7y = 0$ (d) $7x - 3y = 0$
- $\tan \theta \operatorname{cosec}^2 \theta - \tan \theta$ is equal to
 (a) $\sec \theta$ (b) $\cot^2 \theta$ (c) $\sin \theta$ (d) $\cot \theta$
- If the ratio of the height of a tower and the length of its shadow is $\sqrt{3}:1$, then the angle of elevation of the sun has measure :
 (a) 45° (b) 30° (c) 90° (d) 60°
- The height of a right circular cone whose radius is 5 cm and slant height is 13 cm will be
 (a) 12 cm (b) 10 cm (c) 13 cm (d) 5 cm
- Which of the following is incorrect ?
 (a) $P(A) > 1$ (b) $0 \leq P(A) \leq 1$ (c) $P(\emptyset) = 0$ (d) $P(A) + P(\bar{A}) = 1$
- The range of the data 8, 8, 8, 8, ... 8, is :
 (a) 0 (b) 1 (c) 8 (d) 3

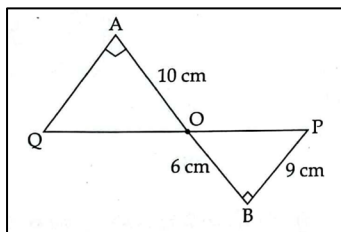
PART - II

Note : Answer any 10 questions. Question No. 28 is compulsory.

 $10 \times 2 = 20$

- If $B \times A = \{(-2,3), (-2,4), (0,3), (0,4), (3,3), (3,4)\}$ find A and B.
- Given $f(x) = 2x - x^2$. Find (i) $f(1)$ (ii) $f(x+1)$.
- Find the 8th term of the G.P. 9, 3, 1,

18. Find the LCM of $9a^3b^2$, $12a^2b^2c$.
19. Determine the nature of the roots for the following quadratic equation $15x^2 + 11x + 2 = 0$.
20. In figure QA and PB are perpendicular to AB. If $AO = 10\text{ cm}$, $BO = 6\text{ cm}$ and $PB = 9\text{ cm}$. Find AQ.



21. The line p passes through the points $(3, -2)$, $(12, 4)$ and the line q passes through the points $(6, -2)$ and $(12, 2)$. Is p parallel to q ?
22. Find the slope of the straight line $6x + 8y + 7 = 0$.
23. Prove the following identity $\frac{1 - \tan^2 \theta}{\cot^2 \theta - 1} = \tan^2 \theta$.
24. A cylindrical drum has a height of 20 cm and base radius of 14 cm. Find its curved surface Area?
25. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of heights.
26. Find the standard deviation of first 21 natural numbers.
27. A Coin is tossed thrice. What is the Probability of getting two Consecutive tails?
28. Which term of an A.P. 16, 11, 6, 1, ... is -54 ?

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime number. Verify $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
30. A function $f: [-5, 9] \rightarrow \mathbb{R}$ is defined as follows:
- $$f(x) = \begin{cases} 6x + 1; & -5 \leq x < 2 \\ 5x^2 - 1; & 2 \leq x < 6 \\ 3x - 4; & 6 \leq x \leq 9 \end{cases}$$
- Find (i) $f(-3) + f(2)$ (ii) $f(7) - f(1)$ (iii) $2f(4) + f(8)$ (iv) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$
31. Find the sum to n terms of the series $3 + 33 + 333 + \dots$
32. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of ' a ' and ' b '
33. Rekha has 15 square Colour papers of sizes 10 cm, 11 cm, 12 cm, ... 24 cm. How much area can be decorated with these Colour papers.
34. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, and $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ then show that $(AB)^T = B^T A^T$.
35. State and prove Pythagoras Theorem.
36. Find the area of the quadrilateral whose vertices are at $(-9, 0)$, $(-8, 6)$, $(-1, -2)$ and $(-6, -3)$.
37. Find the equation of a straight line through the intersection of lines $7x + 3y = 10$, $5x - 4y = 1$ and parallel to the line $13x + 5y + 12 = 0$.
38. A pole 5 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point 'A' on the ground is 60° and the angle of depression to the point 'A' from the top of the tower is 45° . Find the height of the tower. ($\sqrt{3} = 1.732$).
39. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm. How many small spheres can be obtained ?

40. A teacher asked the students to complete 60 pages of a record note book. Eight students have completed only 32, 35, 37, 30, 33, 36, 35 and 37 pages. Find the standard deviation of the pages completed by them.
41. Two dice are rolled. Find the probability that the sum of outcomes is
 (i) equal to 4 (ii) greater than 10 (iii) less than 13.
42. The internal and external diameters of a hollow hemispherical vessel are 20 cm and 28 cm respectively. Find the cost to paint the vessel all over at ₹0.14 per.cm².

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.

- (a) Draw the two tangents from a point which is 10 cm away from the centre of a circle of radius 5 cm. Also, measure the lengths of the tangents.

OR

- (b) Construct a ΔPQR in which $QR = 5$ cm, $\angle P = 40^\circ$ and the median PG from P to, QR is 4.4 cm Find the length of the altitude from P to QR .

44.

- (a) The following table shows the data about the number of pipes and the time taken to fill the same tank.

No. of Pipes (x)	2	3	6	9
Time Taken (y) (in in)	45	30	15	10

Draw the graph for the above data and hence:

- (i) Find the time taken to fill the tank when five pipes are used.
 (ii) Find the number of pipes when the time is 9 minutes.

OR

- (b) Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.



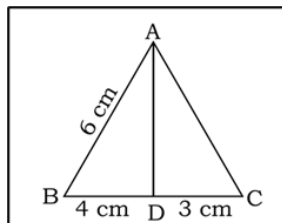
10TH MATHS PUBLIC EXAM APRIL - (2024-2025)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- The range of the relation $R = \{ (x, x^2) | x \text{ is a prime number less than } 33 \}$ is
 (a) $\{2, 3, 5, 7\}$ (b) $\{2, 3, 5, 7, 31\}$
 (c) $\{4, 9, 25, 49, 121\}$ (d) $\{1, 4, 9, 25, 49, 121\}$
- Let $n(A) = m$ and $n(B) = n$, then the total number of functions that exist from B to A is
 (a) m^n (b) n^m (c) $2^{mn} - 1$ (d) 2^{mn}
- Euclid's division lemma states & at for positive integers a and b , there exist unique positive integers q and r such that $a = bq + r$, where x must satisfy.
 (a) $1 < r < b$ (b) $0 < r < b$ (c) $0 \leq r < b$ (d) $3 < r \leq b$
- If 6 times of 6th term of an A.P. is equal to 7 times the 7th term, then the 13th term of the A.P. is:
 (a) 0 (b) 6 (c) 7 (d) 13
- If $(x - 6)$ is the H.C. F of $x^2 - 2x - 24$ and $x^2 - kx - 6$, then the value of k is :
 (a) 3 (b) 5 (c) 6 (d) 8
- Transpose of a column matrix is :
 (a) unit matrix (b) diagonal matrix (c) column matrix (d) row matrix
- A tangent is perpendicular to the radius of a circle at the :
 (a) centre (b) Point of Contact (c) Infinity (d) Chord
- The equation of a line passing through the origin and parallel to the line $7x - 3y + 4 = 0$ is
 (a) $7x - 3y + 4 = 0$ (b) $3x - 7y + 4 = 0$
 (c) $3x + 7y = 0$ (d) $7x - 3y = 0$
- The slope of the line which is perpendicular to a line joining the points $(0,0)$ and $(-8,8)$ is
 (a) -1 (b) 1 (c) $\frac{1}{3}$ (d) -8
- If the ratio of the height of a tower the length of its shadow is $\sqrt{3}:1$, then the angle of elevation of the sun has measure :
 (a) 45° (b) 30° (c) 90° (d) 60°
- The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is
 (a) $60 \pi \text{ cm}^2$ (b) $68 \pi \text{ cm}^2$ (c) $120 \pi \text{ cm}^2$ (d) $136 \pi \text{ cm}^2$
- The volume (in cm^3) of the greatest sphere that can be cut off from a cylindrical log of wood of base radius 1 cm and height 5 cm is :
 (a) $\frac{4}{3} \pi$ (b) $\frac{10}{3} \pi$ (c) 5π (d) $\frac{20}{3} \pi$
- Which of the following is incorrect?
 (a) $P(A) > 1$ (b) $0 \leq P(A) \leq 1$ (c) $P(\varphi) = 0$ (d) $P(A) + P(\bar{A}) = 1$
- A purse contains 10 notes of ₹ 2000, 15 notes of ₹ 500 and 25 notes of ₹ 200. One note is drawn at random. What is the probability that the note is either a ₹ 500 note or ₹ 200 note?
 (a) $\frac{1}{5}$ (b) $\frac{3}{10}$ (c) $\frac{2}{3}$ (d) $\frac{4}{5}$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- Let $A = \{1, 2, 3\}$ and $B = \{x \mid x \text{ is a prime number less than } 10\}$. Find $A \times B$.
- Find $f \circ g$ and $g \circ f$ when $f(x) = 2x + 1$ and $g(x) = x^2 - 2$.

17. A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over.
18. Add : $\frac{x^3}{x-y} + \frac{y^3}{y-x}$.
19. If $A = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ find AB and BA . Verify if $AB = BA$.
20. In the given diagram, AD is the bisector of $\angle A$. If $BD = 4$ cm, $DC = 3$ cm and $AB = 6$ cm, Find AC .



21. The line through the points $(-2, a)$ and $(9, 3)$ has slope $-\frac{1}{2}$. Find the value of 'a'.
22. Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular.
23. From the top of a rock $50\sqrt{3}$ m high, the angle of depression of a car on the ground is observed to be 30° . Find the distance of the car from the rock.
24. If the ratio of radii of two spheres is $4 : 7$, find the ratio of their volumes.
25. The slant height of a frustum of a cone is 5 cm and the radii of its ends are 4 cm and 1 cm. Find its curved surface area.
26. Find the range of the following distribution :
- | | | | | | | |
|--------------------|---------|---------|---------|---------|---------|---------|
| Age (in Years) | 16 – 18 | 16 – 18 | 16 – 18 | 16 – 18 | 16 – 18 | 16 – 18 |
| Number of Students | 0 | 4 | 6 | 8 | 2 | 2 |
27. A flower is selected at random from a basket containing 80 yellow, 70 red and 50 white flowers. Find the probability of selecting a yellow coloured flower.
28. Check whether the sequence $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \sqrt{50}, \dots$ is an A.P or not?. If it is an A.P, then find the common difference.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8, C = The set of even prime numbers. Verify that $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
30. Let $f: A \rightarrow B$ be a function defined by $f(x) = \frac{x}{2} - 1$, where $A = \{2, 4, 6, 10, 12\}$, $B = \{0, 1, 2, 4, 5, 9\}$. Represent f by
 (i) set of ordered pairs (ii) an arrow diagram (iii) a table (iv) a graph
31. If $p_1^{x_1} \times p_2^{x_2} \times p_3^{x_3} \times p_4^{x_4} = 113400$ where p_1, p_2, p_3, p_4 are primes in ascending order and x_1, x_2, x_3, x_4 are integers, find the value of p_1, p_2, p_3, p_4 and x_1, x_2, x_3, x_4 .
32. Find the sum of the series $6^2 + 7^2 + 8^2 + \dots + 21^2$.
33. If $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 4 \\ 0 & 2 \end{pmatrix}$, show that $(AB)^T = B^T A^T$.
34. State and prove Angle Bisector Theorem.
35. Find the value of k , if the area of a quadrilateral is 28 sq. units, whose vertices are taken in the order $(-4, -2), (-3, k), (3, -2)$ and $(2, 3)$.
36. A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3, 8)$. Find its equation.

37. Prove that: $\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$.
38. Nathan, an engineering student was asked to make a model shaped like a cylinder with two cones attached at its two ends. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of the model that Nathan made.
39. A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm. How many small spheres can be obtained?
40. The total marks scored by two students, Sathya and Vidhya in 5 subjects are 460 and 480 with standard deviation 4.6 and 2.4 respectively. Who is more consistent in performance?
41. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn from the bag. Find the probability that the ball drawn is
 (i) white (ii) black or red (iii) not white (iv) neither white nor black
42. Determine the nature of the roots of the equation
 $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.
 (a) Draw a triangle ABC base $BC = 8$ cm, $\angle A = 60^\circ$ and the bisector of $\angle A$ meets BC at D such that $ED = 6$ cm.
OR
 (b) Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw the two tangents to the circle from that point.
44.
 (a) Graph the quadratic equation $x^2 - 9x + 20 = 0$ and state the nature of solution.
OR
 (b) A bus is travelling at a uniform speed of 50 km/hr. Draw the distance - time graph and hence find.
 (i) the constant of variation.
 (ii) how far will it travel in 90 minutes?
 (iii) The time required to cover a distance of 300 km.



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10TH MATHS PUBLIC EXAM APRIL - (2025-2026)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- If $\{(a, 8)(6, b)\}$ represents an identity function, then the value of a and b are respectively:
(a) (8,6) (b) (8,8) (c) (6,8) (d) (6,6)
- $f(x) = (x + 1)^3 - (x - 1)^3$ represents a function which is :
(a) linear (b) cubic (c) reciprocal (d) quadratic
- If the H.C.F of 65 and 117 is expressible in the form of $65m - 117$, then the value of m is:
(a) 4 (b) 2 (c) 1 (d) 3
- The value of $(1^3 + 2^3 + 3^3 + \dots + 15^3) - (1 + 2 + 3 + \dots + 15)$ is:
(a) 14400 (b) 14200 (c) 14280 (d) 14520
- The solution of the system $x + y - 3z = -6$, $-7y + 7z = 7$, $3z = 9$ is:
(a) $x = 1, y = 2, z = 3$ (b) $x = -1, y = 2, z = 3$
(c) $x = -1, y = 2, z = 3$ (d) $x = 1, y = -2, z = 3$
- For the given matrix $A = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \\ 9 & 11 & 13 & 15 \end{pmatrix}$, the order of the matrix A^T is:
(a) 2×3 (b) 3×2 (c) 3×4 (d) 4×3
- Two poles of heights 6 m and 11 m stand vertically on a plane ground. If the distance between their feet is 12 m, what is the distance between their tops?
(a) 13 m (b) 14 m (c) 15 m (d) 12.8 m
- The point of intersection of the straight lines $3x - y = 4$ and $x + y = 8$ is:
(a) (5, 3) (b) (2, 4) (c) (3,5) (d) (4, 4)
- The straight line given by the equation $x = 11$ is:
(a) Parallel to X axis (b) Parallel to Y axis
(c) Passing through the origin (d) Passing through the point (0, 11)
- If $\tan\theta + \cot\theta = 2$, then $\tan^2\theta + \cot^2\theta$ is equal to :
(a) 2 (b) 4 (c) $\sqrt{2}$ (d) 0
- A solid sphere of radius x cm is melted and cast into a shape of a solid cone of same radius. The height of the cone is :
(a) $3x$ cm (b) x cm (c) $4x$ cm (d) $2x$ cm
- A spherical ball of radius r_1 units is melted to make 8 new identical balls each of radius r_2 units. Then $r_1:r_2$ is:
(a) 2: 1 (b) 1: 2 (c) 4: 1 (d) 1: 4
- For a collection of data, $\Sigma x^2 = 140$, $\Sigma x = 28$ and $n = 7$, the standard deviation is:
(a) 4 (b) 2 (c) 16 (d) 8

The probability of a red marble selected at random from a jar containing p red, q blue and r green marbles is: (a) $\frac{q}{p+q+r}$ (b) $\frac{p}{p+q+r}$ (c) $\frac{p+q}{p+q+r}$ (d) $\frac{p+r}{p+q+r}$

PART - II

Note : Answer any 10 questions. Question No. 28 is compulsory.

10 × 2 = 20

15. A relation R is given by the set $\{(x, y)/y = x + 3, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and range.
16. Let $X = \{-5, 1, 3, 4\}$ and $Y = \{a, b, c\}$. Determine whether the relation $R = \{(-5, a)(1, a)(3, b)\}$ is a function from X to Y ?
17. Find the least positive value of x such that $71 \equiv x \pmod{8}$
18. Find the sum to infinity of $9 + 3 + 1 + \dots$
19. Reduce the rational expression $\frac{x^2-16}{x^2+8x+16}$ to its lowest form.
20. If the difference between a number and its reciprocal is $\frac{24}{5}$ find the number.
21. Construct a 3×3 matrix whose elements are given by $a_{ij} = |i - 2j|$
22. D and E are respectively the points on the sides AB and AC of a $\triangle ABC$ such that $AB = 5.6$ cm, $AD = 1.4$ cm, $AC = 7.2$ cm and $AE = 1.8$ cm, show that $DE \parallel BC$.
23. Find the slope of a line joining the points $(14, 10)$ and $(14, -6)$.
24. Find the equation of a straight line whose intercepts on the x and y axes are 4 and -6 respectively.
25. Prove that $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \operatorname{cosec}\theta + \cot\theta$
26. Find the Standard Deviation of first 21 natural numbers.
27. A die is rolled and a coin is tossed simultaneously. Find the Probability that the die shows an odd number and the coin shows a head.
28. Find the ratio of the volume of a cube to that of a largest sphere which exactly fits into the cube.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{x \in W/x < 2\}$, $B = \{x \in N/1 < x \leq 4\}$ and $C = \{3, 5\}$.
Verify that $A \times (B \cap C) = (A \times B) \cap (A \times C)$
30. If $f(x) = x - 1$, $g(x) = 3x + 1$ and $h(x) = x^2$, then prove that $(f \circ g) \circ h = f \circ (g \circ h)$.
31. Find the sum of all natural numbers between 300 and 600 which are divisible by 7.
32. Find the sum to n terms of the series $3 + 33 + 333 + \dots$

33. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$ verify that $(AB)^T = B^T A^T$.
34. Show that the angle bisectors of a triangle are concurrent.
35. Find the area of the quadrilateral formed by the points (8,6), (5,11), (−5,12) and (−4,3).
36. $A(-3,0)$, $B(10,-2)$ and $C(12,3)$ are the vertices of $\triangle ABC$. Find the equation of the altitude through A.
37. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between two ships. ($\sqrt{3} = 1.732$)
38. A garden roller whose length is 3m long and whose diameter is 2.8m is rolled to level a garden. How much area will it cover in 8 revolutions?
39. A solid right circular cone of diameter 14 cm and height 8 cm is melted to form a hollow sphere. If the external diameter of the sphere is 10 cm, find the internal diameter.
40. Find the coefficient of variation of 24,26,33,37,29,31.
41. Two dice are rolled together. Find the Probability of getting a doublet or sum of faces as 4.
42. If α and β are the roots of the equation $2x^2 - 3x + 1 = 0$, then form a quadratic equation whose roots are $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$

PART - IV

Note : Answer the following questions..

$2 \times 8 = 16$

43. (a) Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR. (Scale factor $\frac{3}{5} < 1$).
- OR**
- (b) Draw a circle of diameter 6 cm from a point P, which is 8 cm away from its centre. Draw two tangents PA and PB to the circle and measure their lengths.
44. (a) Graph the quadratic equation $x^2 - 6x + 9 = 0$ and state its nature of solutions.
- OR**
- (b) Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find y when $x = 9$ (ii) find x when $y = 7.5$.



10TH MATHS PUBLIC EXAM JULY - (2024-2025)

Time Allowed: 3.00 Hours

Maximum Marks: 100

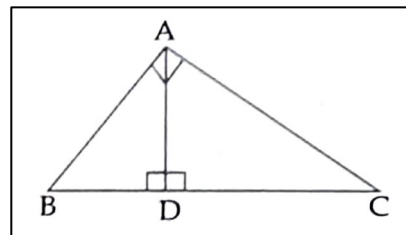
PART - I

Note : (i) Answer all the questions.

14 × 1 = 14

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- $A = \{a, b, p\}$, $B = \{2, 3\}$, $C = \{p, q, r, s\}$ then $n[(A \cup C) \times B]$ is:
(a) 8 (b) 20 (c) 12 (d) 16
- If $g = \{(1, 1), (2, 3), (3, 5), (4, 7)\}$ is a function given by $g(x) = \alpha x + \beta$, then the values of α and β are:
(a) $(-1, 2)$ (b) $(2, -1)$ (c) $(-1, -2)$ (d) $(1, 2)$
- The value of $(1^3 + 2^3 + 3^3 + \dots + 15^3) - (1 + 2 + 3 + \dots + 15)$ is:
(a) 14400 (b) 14200 (c) 14280 (d) 14520
- The n^{th} term of an A.P. is $2n - 1$. The sum of the first 'n' terms of that A.P. is :
(a) n^2 (b) $2n - 1$ (c) $n + 2$ (d) $n(n + 2)$
- The solution of the system $x + y - 3z = -6$, $-7y + 7z = 7$, $3z = 9$ is
(a) $x = 1, y = 2, z = 3$ (b) $x = -1, y = 2, z = 3$
(c) $x = -1, y = -2, z = 3$ (d) $x = 1, y = -2, z = 3$
- If A is a 2×3 matrix and B is a 3×4 matrix how many columns does AB have?
(a) 3 (b) 4 (c) 2 (d) 5
- In the given figure $\angle BAC = 90^\circ$ and $AD \perp BC$ then
(a) $BD \cdot CD = BC^2$ (b) $AB \cdot AC = BC^2$
(c) $BD \cdot CD = AD^2$ (d) $AB \cdot AC = AD^2$
- The area of a triangle formed by the points $(-5, 0)$, $(0, -5)$ and $(5, 0)$ is
(a) 0 sq. units (b) 25 sq. units (c) 5 sq. units (d) none of these
- When proving that a quadrilateral is a trapezium, it is necessary to show
(a) Two sides are parallel (b) Two parallel and two non-parallel sides
(c) Opposite sides are parallel (d) An sides are of equal length
- The value of $\sin^2 \theta + \frac{1}{1 + \tan^2 \theta}$ is equal to :
(a) $\tan^2 \theta$ (b) 1 (c) $\cot^2 \theta$ (d) 0
- The curved surface area of a right circular cone of height 15cm base diameter 26 cm
(a) $60\pi \text{ cm}^2$ (b) $68\pi \text{ cm}^2$ (c) $120\pi \text{ cm}^2$ (d) $136\pi \text{ cm}^2$
- The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is
(a) 1:2:3 (b) 2:1:3 (c) 1:3:2 (d) 3:1:2
- A page is selected at random from a book. The probability that the digit at units place of the page number chosen is less than 7 is :
(a) $\frac{3}{10}$ (b) $\frac{7}{10}$ (c) $\frac{3}{9}$ (d) $\frac{7}{9}$
- If the standard deviation of a data is $\sqrt{2}$ and each value is increased by 2, then the new standard deviation of that data is
(a) $2 + \sqrt{2}$ (b) $2\sqrt{2}$ (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

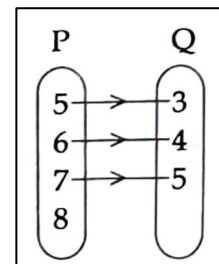


PART - II

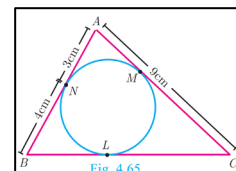
Note : Answer any 10 questions. Question No. 28 is compulsory.

10 × 2 = 20

15. If $A = \{2, -2, 3\}$, $B = \{1, -4\}$, then find $A \times B, B \times A$.
16. The arrow diagram shows the relationship between the sets P and Q . Write the relation in
 (i) Set builder form (ii) Roster form
17. 'a' and 'b' are two positive integers such that $a^b \times b^a = 800$. Find 'a' and 'b'.
18. In a G.P. 729, 243, 81 ... find t_7



19. Find the excluded values of the expression $\frac{t}{t^2 - 5t + 6}$, if any.
20. The slant height of a frustrum of a cone is 5 cm and radii of its ends are 4 cm and 1 cm Find its curved surface area.
21. In the given figure, ΔABC is circumscribing a circle. Find the length of BC .
22. Find the slope of a line joining the points $(-6, 1)$ and $(-3, 2)$.
23. Find the intercepts made by the line $4x + 3y + 12 = 0$ on the coordinate axes.
24. Prove $\frac{\cos \theta}{1 + \sin \theta} = \sec \theta - \tan \theta$
25. The curved surface area of a right circular cylinder of height 14 cm is 88 cm^2 . Find its diameter.
26. If the range and the smallest value of a set of data are 36.8 and 13.4 respectively, then find The largest value.
27. Two coins are tossed together. What is the probability of getting different faces on the coins?
28. Given that $A = \begin{pmatrix} -1 & 1 \\ a & b \end{pmatrix}$ where $a, b \geq 0$ find the values of a and b if $A^2 = I$.



PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5, 8, 11, 14\}$ be two sets. Let $f: A \rightarrow B$ be a function given by $f(x) = 3x - 1$. Represent this function
 (i) by arrow diagram (ii) in a table form
 (iii) as a set of ordered pairs (iv) in a graphical form
30. If $f(x) = x - 1$, $g(x) = 3x + 1$ and $h(x) = x^2$, then show that $(f \circ g) \circ h = f \circ (g \circ h)$
31. Find the H.C.F of 396, 504, 636.
32. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm, ..., 24 cm. How much area can be decorated with these colour papers?
33. Find the values of m and n of the polynomial $36x^4 - 60x^3 + 61x^2 - mx + n$ are perfect squares.
34. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ show that $A^2 - 5A + 7I_2 = 0$.
35. State and prove Pythagoras Theorem.
36. Find the area of the quadrilateral formed by the points $(-9, -2)$, $(-8, -4)$, $(2, 2)$ and $(1, -3)$.
37. From the top of a 12 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 30° . Determine the height of the tower.
38. A toy is in the shape of a cylinder surmounted by a hemisphere. The height of the toy is 25 cm. Find the total surface area of the toy if its common diameter is 12 cm.
39. An Aluminum sphere of radius 12 cm is melted to make a cylinder of radius 8 cm. Find the height of the cylinder.
40. The marks scored by the students in a slip test are given below. Find the standard deviation of their marks.

x	4	6	8	10
f	7	3	5	9

41. Two dice are rolled. Find the probability that the sum of outcomes is
 (i) equal to 4 (ii) greater than 10 (iii) less than 13
42. Find the Equation of a straight line perpendicular to $2x - 3y + 6 = 0$ and passes through the midpoint of the line segment joining the points $(-4, -7)$ and $(6, 5)$.

PART - IV**Note : Answer the following questions..** **$2 \times 8 = 16$**

43.

- (a) Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{7}{4}$ of the corresponding sides of the triangle PQR. (scale factor $\frac{7}{4} > 1$)

OR

- (a) Draw two tangents from a point which is 5 cm away from the centre of a circle of diameter 6 cm. Also, measure the lengths of the tangents.

44.

- (a) Graph the quadratic equation $x^2 + x - 12 = 0$ and determine the nature of its roots.

OR

- (b) A school announces that for a certain competition the cash prize will be distributed for all the participants equally as shown below :

No .of Participants x	2	4	6	8	10
Amount for each participant y in ₹	180	90	60	45	36

- (i) Find the constant of variation.
- (ii) Graph the above data and hence find how much each participant will get if the number of participants are 12.

**PREPARED & TYPED BY****Y.SEENIVASAN. M.Sc. B.Ed****MATHS - TEACHER -8489880553**

10TH MATHS PUBLIC EXAM JULY - (2025-2026)**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- If there are 1024 relations from a set $A = \{1, 2, 3, 4, 5\}$ to a set B, then the number of elements in B is:
(a) 3 (b) 2 (c) 4 (d) 8
- If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$ then $(f \circ g)(x)$ is:
(a) $\frac{3}{2x^2}$ (b) $\frac{2}{3x^2}$ (c) $\frac{2}{9x^2}$ (d) $\frac{1}{6x^2}$
- Given $F_1 = 1, F_2 = 3$ and $F_n = F_{n-1} + F_{n-2}$ then F_5 is:
(a) 3 (b) 5 (c) 8 (d) 11
- If 6 times 6th term of an A.P. is equal to 7 times the 7th term, then the 13th term of the A.P. is:
(a) 0 (b) 6 (c) 7 (d) 13
- $\frac{a^2}{a-b} + \frac{b^2}{b-a}$ is equal to :
(a) $\frac{a^2+b^2}{a-b}$ (b) $a^2 - b^2$ (c) $a + b$ (d) $a - b$
- Find the matrix X if $2X + \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 5 & 7 \\ 9 & 5 \end{pmatrix}$
(a) $\begin{pmatrix} -2 & -2 \\ 2 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix}$ (c) $\begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$ (d) $\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$
- How many tangents can be drawn to the circle from an exterior point ?
(a) One (b) Two (c) Infinite (d) Zero
- The slope of the line joining (12, 3) and (4, a) is $\frac{1}{8}$. The value of 'a' is:
(a) 1 (b) 4 (c) -5 (d) 2
- A man walks near a wall, such that the distance between him and the wall is 10 units. Consider the wall to be the Y axis. The path travelled by the man is :
(a) $x = 10$ (b) $y = 10$ (c) $x = 0$ (d) $y = 0$
- If $x = a \tan \theta$ and $y = b \sec \theta$ then:
(a) $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (c) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (d) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$
- If the volume of solid hemisphere is numerically equal to its total surface area then the diameter is:
(a) $\frac{9}{2}$ units (b) 9 units (c) 3π units (d) $\frac{3}{2}$ units
- The total surface area of a cylinder whose radius is $\frac{1}{3}$ of its height is:
(a) $\frac{9\pi h^2}{8}$ sq. units (b) $24\pi h^2$ sq. units (c) $\frac{8\pi h^2}{9}$ sq. units (d) $\frac{56\pi h^2}{9}$ sq. units
- The sum of all deviations of the data from its mean is:
(a) Always positive (b) Always negative
(c) Zero (d) Non-zero integer
- A page is selected at random from a book. The probability that the digit at units place of the Page number chosen is less than 7 is:
(a) $\frac{3}{10}$ (b) $\frac{7}{10}$ (c) $\frac{3}{9}$ (d) $\frac{7}{9}$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- If $B \times A = \{(-2, 3), (-2, 4), (0, 3), (0, 4), (3, 3), (3, 4)\}$ find A and B.
- If $f(x) = x^2 - 1, g(x) = x - 2$ find a, if $(g \circ f)(a) = 1$

17. Find the terms a_6 and a_{13} of the sequence whose n^{th} term is given by $a_n = \frac{5n}{n+2}$
18. Find the sum of: $1 + 3 + 5 + \dots + 55$
19. Determine the quadratic equation, whose sum and product of the roots are $-9, 20$ respectively.
20. Simplify: $\frac{x^2-16}{x+4} \div \frac{x-4}{x+4}$
21. Verify that $A^2 = I$ when $A = \begin{pmatrix} 5 & -4 \\ 6 & -5 \end{pmatrix}$
22. If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3$ cm, $EF = 4$ cm and area of $\triangle ABC = 54$ cm², find the area of $\triangle DEF$.
23. Find the area of the triangle whose vertices are $(-3, 5)$, $(5, 6)$ and $(5, -2)$
24. Calculate the slope and y intercept of the straight line $8x - 7y + 6 = 0$
25. A cylindrical drum has a height of 20 cm and base radius of 14 cm. Find its curved surface area.
26. The standard deviation and mean of a data are 6.5 and 12.5 respectively. Find the coefficient of variation.
27. What is the probability that a leap year selected at random will contain 53 Saturdays ?
28. Prove that $\sqrt{\frac{1+\tan^2\theta}{1+\cot^2\theta}} = \tan\theta$

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{x \in W | x < 2\}$, $B = \{x \in N | 1 < x \leq 4\}$ and $C = \{3, 5\}$.
Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$
30. Let f be a function $f: N \rightarrow N$ be defined by $f(x) = 3x + 2$, $x \in N$
(i) Find the images of 1, 2, 3. (ii) Find the pre-images of 29, 53.
(iii) Identify the type of function.
31. A mother divides into three parts such that the amounts are in A.P. and gives it to her three children. The product of the two least amounts that the children had is 4,623. Find the amount received by each child.
32. Find the sum of the Geometric series: $3 + 6 + 12 + \dots + 1536$
33. Find the square root of the polynomial $37x^2 - 28x^3 + 4x^4 + 42x + 9$ by division method.
34. Find the matrix X and Y if $X + Y = \begin{bmatrix} 7 & 0 \\ 3 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$
35. State and prove Pythagoras theorem.
36. If the vertices of a $\triangle ABC$ are $A(6, 2)$, $B(-5, -1)$ and $C(1, 9)$ then find the equation of median through A .
37. An aeroplane at an altitude of 1800 m finds that two boats are sailing towards it in the same direction. The angles of depression of the boats as observed from the aeroplane are 60° and 30° respectively. Find the distance between the two boats. ($\sqrt{3} = 1.732$)
38. If the circumference of a conical wooden piece is 484 cm then find its volume when its height is 105 cm.
39. A capsule is in the shape of a cylinder with two hemisphere stuck to each of its ends. If the length of the entire capsule is 12 mm and the diameter of the capsule is 3 mm, how much volume of medicine can it hold ?
40. Find the mean and variance of the first 'n' natural numbers.
41. Three fair coins are tossed together. Find the probability of getting:
(i) all heads (ii) at least one tail
(iii) at most one head (iv) at most two tails.
42. A straight line $7x + 16y - 56 = 0$ meets the X and Y axis at A and B respectively. Find the area of $\triangle OAB$ where O is the origin.

PART - IV

Note : Answer the following questions..

$2 \times 8 = 16$

45.

- (b) Construct a ΔPQR where the base $PQ = 4.5$ cm, $\angle R = 35^\circ$ and the median RG from R to PQ is 6 cm.

OR

- (c) Draw a tangent to the circle from the point P having radius 3.6 cm, and centre at O , point P is at a distance 7.2 cm from the centre.

46.

- (b) Draw the graph of $y = x^2 + 3x - 4$ and hence use it to solve $x^2 + 3x - 4 = 0$

OR

- (d) A garment shop announces a flat 50% discount on every purchase of items for their customers.

Draw the graph for the relation between the Marked price and the Discount. Hence find:

- (i) the marked price when a customer gets a discount of ₹ 3,250 (from graph).
(ii) the discount when the marked price is ₹ 2,500.



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10TH MATHS PTA QUESTIONS COLLECTIONS10TH MATHS PTA QUESTION MODEL -1

Time Allowed: 3.00 Hours

Maximum Marks: 100

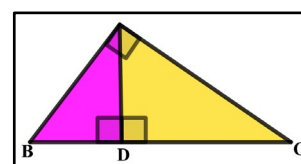
PART - I

Note : (i) Answer all the questions.

14 × 1 = 14

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If $\{(a, 8), (6, b)\}$ represents an identity function, then the value of a and b are respectively
1) (6, 8) 2) (8, 6) 3) (8, 8) 4) (6, 6)
- $7^{4k} \equiv \underline{\hspace{1cm}} \pmod{100}$.
1) 4 2) 3 3) 2 4) 1
- A system of three linear equations in three variables is inconsistent if their planes
1) intersect only at a point 2) intersect in a line
3) coincides with each other 4) do not intersect
- In the adjacent figure $\angle BAC = 90^\circ$ and $AD \perp BC$ then,
1) $BD \cdot CD = BC^2$ 2) $AB \cdot AC = BC^2$
3) $BD \cdot CD = AD^2$ 4) $AB \cdot AC = AD^2$
- The straight line given by the equation $x = 11$
1) Passing through the origin 3) Parallel to X -axis
2) Passing through the point (0,11) 4) Parallel to Y -axis
- If $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\cos \alpha + \sec \alpha)^2 = k + \tan^2 \alpha + \cot^2 \alpha$ then the value of k is equal to
1) 3 2) 5 3) 7 4) 9
- The total surface area of a cylinder whose radius is $1/3$ of its height is
1) $\frac{8\pi h^2}{9}$ sq.units 2) $\frac{9\pi h^2}{8}$ sq. units 3) $\frac{56\pi h^2}{9}$ sq. units 4) $24\pi h^2$ sq. units
- Which of the following is incorrect?
1) $P(A) + P(\bar{A}) = 1$ 2) $P(\phi) = 0$ 3) $0 \leq P(A) \leq 1$ 4) $P(A) > 1$
- The sequence $-3, -3, -3$ is
1) an A.P. only 2) a G.P. only
3) neither A.P nor G.P 4) both A.P and G.P
- The L.C. M of $x^3 - a^3$ and $(x - a)^2$ is
1) $(x^3 - a^3)(x + a)$ 2) $(x^3 - a^3)(x - a)^2$
3) $(x - a)^2(x^2 + ax + a^2)$ 4) $(x + a)^2(x^2 + ax + a^2)$
- In $n(A) = p, n(B) = q$ then the total number of relations that exists between A and B is
1) 2^p 2) 2^q 3) 2^{p+q} 4) 2^{pq}
- If the HCF of 65 and 117 is expressible in the form of $65m - 117$ then, the value of m is
1) 1 2) 3 3) 2 4) 4
- The sum of all deviations of the data from its mean is
1) always positive 2) always negative 3) zero 4) non-zero integer
- The angle of elevation and depression are usually measured by a device called
1) Theodolite 2) Kaleidoscope 3) Periscope 4) Telescope



PART - II

Note : Answer any 10 questions. Question No. 28 is compulsory.

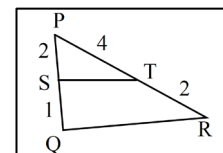
10 × 2 = 20

- A man has 532 flower pots. He wants to arrange them in rows such that each row contains 21 flower pots. Find the number of completed rows and how many flower pots are left over.
- Solve $x^4 - 13x^2 + 42 = 0$.
- If A is of order $p \times q$ and B is of order $q \times r$. What is the order of AB and BA ?

18. A relation f is defined by $f(x) = x^2 - 2$ where $x \in \{-2, -1, 0, 3\}$.

(i) List the elements of f

(ii) Is f a function?



19. Show that $\Delta PST \sim \Delta PQR$.

20. A tower stands vertically on the ground. From a point on the ground, which is 48 m away from the foot of the tower, the angle of elevation of the top of the tower is 30° . Find the height of the tower.

21. The volume of a solid right circular cone is 11088 cm^3 . If its height is 24 cm then find the radius of the cone.

22. If $P(A) = \frac{2}{3}$, $P(B) = \frac{2}{5}$, $P(A \cup B) = \frac{1}{3}$ then find $P(A \cap B)$.

23. Find $A \times B$ and $A \times A$ for $A = \{m, n\}$, $B = \emptyset$.

24. Find the middle term(s) of an A.P 9, 15, 21, 27, ... , 183.

25. The product of Kumaran's age (in years) two years ago and his age four years from now is one more than twice his present age. What is his present age?

26. Find the equation of a line passing through the point $(-4, 3)$ and having slope $-\frac{7}{5}$.

27. The standard deviation of 20 observations is $\sqrt{6}$. If each observation is multiplied by 3, find the standard deviation and variance of the resulting observations.

28. An organization plans to plant saplings in 25 streets in a town in such a way that one sapling for the first street, three for the second, nine for the third and so on. How many saplings are needed to complete the work?

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. The function t which maps temperature in Celsius (C) into temperature in Fahrenheit (F) is defined by $t(C) = F$ where $F = \frac{9}{5}C + 32$. Find (i) $t(0)$. (ii) the value of C when $t(C) = 212$. (iii) the temperature when the Celsius value is equal to the Fahrenheit value.

30. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm, ... 24 cm. How much area can be decorated with these colour papers?

31. If $A = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ -4 & 2 \end{pmatrix}$, $C = \begin{pmatrix} -7 & 6 \\ 3 & 2 \end{pmatrix}$ verify that $A(B + C) = AB + AC$.

32. State and prove Pythagoras theorem.

33. As observed from the top of a 60 m high light house from the sea level the angles of depression of two ships are 28° and 45° . If one ship is exactly behind the other on the same side of the light house. Find the distance between the two ships. ($\tan 28^\circ = 0.5317$).

34. Find the number of coins, 1.5 cm in diameter and 2 mm thick, to be melted to form a right circular cylinder of height 10 m and diameter 4.5 cm.

35. The marks scored by the students in a slip test are given below. Find the standard deviation of their marks.

x	4	6	8	10	12
f	7	3	5	9	5

36. Let A = The set of all natural numbers less than 8, B = The set of all prime numbers less than 8. C = The set of all even prime numbers. Verify that $A \times (B - C) = (A \times B) - (A \times C)$.

37. If $S_n = (x + y) + (x^2 + xy + y^2) + (x^3 + x^2y + xy^2 + y^3) + \dots$ n terms prove that

$$(x - y)S_n = \frac{x^2(x^n - 1)}{x - 1} - \frac{y^2(y^n - 1)}{y - 1}.$$

38. Solve $\frac{1}{2x} + \frac{1}{4y} - \frac{1}{3z} = \frac{1}{4}$, $\frac{1}{x} = \frac{1}{3y}$; $\frac{1}{x} - \frac{1}{5y} + \frac{4}{z} = 2\frac{2}{15}$.

39. A funnel consists of a frustrum of a cone attached to a cylindrical portion 12 cm long attached at the bottom. If the total height be 20 cm, diameter of the cylindrical portion be 12 cm and the diameter of the top of the funnel be 24 cm. find the outer surface area of the funnel.
40. In a class of 50 students, 28 opted for NCC, 28 opted for NSS and 10 opted for both NCC and NSS. One of the students is selected at random. Find the probability that
 (i) the student opted for NCC but not NSS. (ii) The student opted for NSS but not NCC.
 (iii) The student opted for exactly one of them.
41. The base of a triangle is 4 cm longer than its altitude. If the area of the triangle is 48 sq.cm. then find its base and altitude.
42. The area of a triangle is 5 sq. units. Two of its vertices are (2,1) and (3, -2). The third vertex lies on the line $y = x + 3$. Find the third vertex.

PART - IV

Note : Answer the following questions..

2 × 8 = 16

43.

- a) Draw the graph of $y = x^2 + x - 2$ and hence use it to solve the equation $x^2 + x - 2 = 0$.

(OR)

- b) Solve: $2x + y + 4z = 15$, $x - 2y + 3z = 13$, $3x + y - z = 2$

44.

- a) Construct a triangle similar to a given triangle ABC with its sides equal to $\frac{6}{5}$ of the corresponding sides of the triangle ABC. (Scale Factor $\frac{6}{5} > 1$)

(OR)

- b) ABC is a triangle with $\angle B = 90^\circ$, BC = 3 cm and AB = 4 cm. D is a point on AC such that AD = 1 cm and E is the midpoint of AB. Join D and E and extend DE to meet CB at F. Find BF.



10TH MATHS PTA QUESTION MODEL -2**Time Allowed: 3.00 Hours****Maximum Marks: 100****PART - I****Note : (i) Answer all the questions.****14 × 1 = 14****(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.**

- If $f: A \rightarrow B$ is a bijective function and if $n(B) = 7$, then $n(A)$ is equal to
1) 1 2) 49 3) 14 4) 7
- If there are 1024 relations from a set $A = \{1, 2, 3, 4, 5\}$ to a set B , then the number of elements in B is
1) 2 2) 3 3) 4 4) 8
- The next term of the sequence $\frac{3}{16}, \frac{1}{8}, \frac{1}{12}, \frac{1}{18}$ is
1) $\frac{2}{3}$ 2) $\frac{1}{24}$ 3) $\frac{1}{27}$ 4) $\frac{1}{81}$
- Which of the following should be added to make $x^4 + 64$ a perfect square?
1) $4x^2$ 2) $8x^2$ 3) $-8x^2$ 4) $16x^2$
- The excluded value of the rational expression $\frac{x^3+8}{x^2-2x-8}$ is
1) 8 2) 2 3) 4 4) 1
- Graph of a linear equation is a
1) straight line 2) circle 3) parabola 4) hyperbola
- A Tangent is perpendicular to the radius at the
1) centre 2) infinity 3) point of contact 4) chord
- The area of triangle formed by the points $(-5, 0)$, $(0, -5)$ and $(5, 0)$ is
1) 0 sq.units 2) 5 Sq.units 3) 25 sq.units 4) none of these
- The point of intersection of $3x - y = 4$ and $x + y = 8$ is
1) (3, 5) 2) (2, 4) 3) (5, 3) 4) (4, 4)
- If $5x = \sec \theta$? and $\frac{5}{x} = \tan \theta$ then, $x^2 - \frac{1}{x^2}$ is equal to
1) 1 2) 5 3) 25 4) $\frac{1}{25}$
- $\frac{\sin(90-\theta)}{\tan \theta} + \frac{\cos(90-\theta) \cos \theta}{\cot \theta} =$
1) $\tan \theta$ 2) 1 3) -1 4) $\sin \theta$
- The height and radius of the cone of which the frustum is a part are h_1 units and r_1 units respectively. Height of the frustum is h_2 units and radius of the smaller base is r_2 units if $h_2: h_1 = 1: 2$ then $r_2: r_1$ is
1) 1: 2 2) 2: 1 3) 1: 3 4) 3: 1
- The range of the first 10 prime numbers is
1) 9 2) 20 3) 27 4) 5
- The average of first n natural numbers is
1) $\frac{n(n+1)}{2}$ 2) $\frac{n}{2}$ 3) $\frac{n+1}{2}$ 4) n

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

- A relation R is given by the set $\{(x, y) / y = x^2 + 3x, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and range.
- If $f(x) = x^2 - 1$, $g(x) = x - 2$ and find a if $g \circ f(a) = 1$.

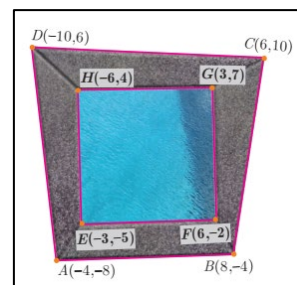
17. If A and B are mutually exclusive events of a random experiment and $P(\text{not } A) = 0.45$, $P(A \cup B) = 0.65$ then find $P(B)$.
18. Dividing the polynomial $P(x) = x^2 - 5x - 14$ by another polynomial $q(x)$ yields $\frac{x-7}{x+2}$ then find $q(x)$.
19. If $\begin{pmatrix} \sqrt{7} & -3 \\ -\sqrt{5} & 2 \\ \sqrt{3} & -5 \end{pmatrix}$ then find the transpose of $-A$.
20. If ΔABC is similar to ΔDEF such that $BC = 3 \text{ cm}$, $EF = 4 \text{ cm}$ and area of $\Delta ABC = 54 \text{ cm}^2$. Find the area of ΔDEF .
21. Find the slope of a line joining the points $(\sin \theta, -\cos \theta)$ and $(-\sin \theta, \cos \theta)$.
22. The hill in the form of a right triangle has its foot at $(19, 3)$. The inclination of the hill to the ground is 45° . Find the equation of the hill joining the foot and top.
23. Find x so that $x + 6$, $x + 12$ and $x + 15$ are three consecutive terms of a Geometric Progression.
24. If $1 + 2 + 3 + \dots + n = 666$ then find n .
25. Find the angle of elevation of the top of a tower from a point on the ground which is 30 m away from the foot of a tower of height $10\sqrt{3} \text{ m}$.
26. The ratio of the radii of two right circular cones of same height is 1:3. Find the ratio of their curved surface area when the height of each cone is 3 times the radius of the smaller cone?
27. If two positive integers p and q are written as $p = a^2b^3$ and $q = a^3b$, a, b are prime numbers then verify $LCM(p, q) \times HCF(p, q) = pq$.
28. Find the number of spherical lead shots, each of diameter 6 cm that can be made from a solid cuboids of lead having dimensions $24 \text{ cm} \times 22 \text{ cm} \times 12 \text{ cm}$.

PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. In the figure, the quadrilateral swimming pool shown is surrounded by concrete patio. Find the area of the patio.
30. State and prove Thales theorem.
31. If $f(x) = x - 4$, $g(x) = x^2$, $h(x) = 3x - 5$ then show that $(f \circ g) \circ h = f \circ (g \circ h)$.
32. (i) Find the least positive value of x such that $67 + x \equiv 1 \pmod{4}$.
(ii) Solve $5x \equiv 4 \pmod{6}$.
33. The houses of a street are numbered from 1 to 49. Senthil's house is numbered such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number.
34. A coin is tossed thrice. Find the probability of getting exactly two heads or atleast one tail or two consecutive heads.
35. The temperature of two cities A and B in a winter season are given below.



Temperature of city A (in degree celsius)	18	20	22	24	26
Temperature of city B (in degree celsius)	11	14	15	17	18

36. Let $A = \{x \in W / x < 2\}$, $B = \{x \in N / 1 \leq x \leq 4\}$ and $C = \{3, 5\}$.
Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
37. Vani her father and her grand father have an average age of 53. One half of her grandfather's age plus one-third of her father's age plus one-fourth of Vani's age is 65. Four years ago if Vani's grandfather was four times as old as Vani then how old are they all now?

38. If $A = \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix}$, $B = \begin{pmatrix} \sin \theta & 0 \\ 0 & \sin \theta \end{pmatrix}$ then show that $A^2 + B^2 = I$.
39. A metallic sheet in the form of a sector of a circle of radius 21 cm has central angle of 216° . The sector is made into a cone by bringing the bounding radii together. Find the volume of the cone formed.
40. A shuttle cock used for playing badminton has the shape of a frustrum of a cone is mounted on a hemisphere. The diameters of the frustrum are 5 cm and 2 cm. The height of the entire shuttle cock is 7 cm. Find its external surface area.
41. A motor Boat whose speed is 18 km / hr in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.
42. A 1.2 m tall girls spots a balloon moving with the wind in a horizontal line at a height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at an instant is 60° . After some time the angle of elevation reduces to 30° . Find the distance travelled by the balloon during the interval.

PART - IV**Note : Answer the following questions..** **$2 \times 8 = 16$**

43.

- a) Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$.

(OR)

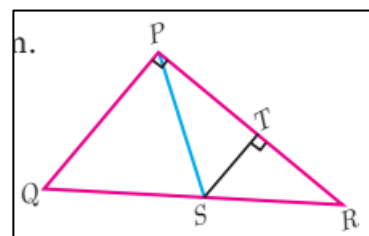
- b) Find the values of a and b if $16x^4 - 24x^3 + (a - 1)x^2 + (b + 1)x + 49$ is a perfect square.

44.

- a) Take a point which is 11 cm away from the centre of a circle of radius 4 cm and draw two tangents to the circle from the point.

(OR)

- b) In a figure $\angle QPR = 90^\circ$. PS is its bisector. If $ST \perp PR$,
Prove that $ST \times (PQ + PR) = PQ \times PR$



10TH MATHS PTA QUESTION MODEL -3

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

$$\boxed{14 \times 1 = 14}$$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- [illegible]

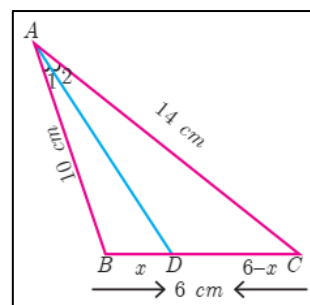
PART - II

Note : Answer **any 10** questions. Question **No. 28** is compulsory.

$$10 \times 2 = 20$$

- 15.** Let f be a function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(x) = 3x + 2, x \in \mathbb{N}$. Find the pre - image of 29, 53.
- 16.** Is $7 \times 5 \times 3 \times 2 \times 3$ a composite number?. Justify your answer.
- 17.** If $3 + k, 18 - k, 5k + 1$ are A.P, then find k .

18. If $1^3 + 2^3 + 3^3 + \dots + k^3 = 16900$ then find $(1 + 2 + 3 + \dots + k)$.
19. If $A = \begin{pmatrix} 7 & 8 & 6 \\ 1 & 3 & 9 \\ -4 & 3 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 11 & -3 \\ -1 & 2 & 4 \\ 7 & 5 & 0 \end{pmatrix}$, then find $2A + B$.
20. If one root of the equation $3x^2 + kx + 81 = 0$ (having real roots) is the square of the other, then find k .
21. If $x = \frac{a^2+3a-4}{3a^2-3}$ and $y = \frac{a^2+2a-8}{2a^2-2a-4}$ find the value of x^2y^{-2} .
22. In the figure, AD is the bisector of $\angle BAC$, if $AB = 10\text{ cm}$, $AC = 14\text{ cm}$ and $BC = 6\text{ cm}$ find BD and DC.
23. What is the inclination of a line whose slope is 1?
24. A player sitting on the top of a tower of height 20 m, observes the angle of depression of a ball lying on the ground as 60° . Find the distance between the foot of the tower and the ball.
($\sqrt{3} = 1.732$).
25. A cone of height 24 cm is made up of modelling clay. A child reshapes it in the form of a cylinder of same radius as cone. Find the height of the cylinder.
26. If A is an event of a random experiment such that $P(A):P(\bar{A}) = 17:15$ and $n(S) = 640$ then find $P(\bar{A})$.
27. The mean of a data is 25.6 and its coefficient of variation is 18.75. find the standard deviation.
28. Show that the straight lines $3x - 5y + 7 = 0$ and $15x + 9y + 4 = 0$ are perpendicular.



PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5, 8, 11, 14\}$ be two sets. Let $f: A \rightarrow B$ be a function given by $f(x) = 3x - 1$. Represent this function
(i) by arrow diagram (ii) in a table form
(iii) as a set of ordered pairs (iv) in a graphical form.
30. An object travels under the influence of gravity in time t seconds is given by $s(t) = \frac{1}{2}gt^2 + at + b$ where (g is the acceleration due to gravity) a, b are the constants. Check if the function $s(t)$ is one - one.
31. Let $A = \{x \in W / 0 < x < 5\}$, $B = \{x \in W / 0 \leq x \leq 2\}$ and $C = \{x \in W / x < 3\}$.
Verify that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
32. Find the sum of the geometric series $3 + 6 + 12 + \dots + 1536$.
33. Find the sum of all 3 digit natural numbers which are divisible by 9.
34. Find the square root of the expression $\frac{4x^2}{y} + \frac{20x}{y} + 13 - \frac{30y}{x} + \frac{9y^2}{x^2}$.
35. Solve the quadratic equation by completing the square method $\frac{5x+7}{x-1} = 3x + 2$.
36. If $A = \begin{pmatrix} 5 & 2 & 9 \\ 1 & 2 & 8 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 7 \\ 1 & 2 \\ 5 & -1 \end{pmatrix}$, verify that $(AB)^T = B^T A^T$.
37. The hypotenuse of a right triangle is 6 m more than twice of the shortest side. If the third side is 2 m less than the hypotenuse. Find the sides of the triangle.
38. Find the equation of a straight line joining the point of intersection of $3x + y + 2 = 0$ and $x - 2y - 4 = 0$ to the point of intersection of $7x - 3y = -12$ and $2y = x + 3$.
39. If $\sqrt{3} \sin \theta - \cos \theta$, then show that $\tan 3\theta = \frac{(3 \tan \theta - \tan^3 \theta)}{1 - 3 \tan^2 \theta}$.

40. The radius of a conical tent is 7 m and the height is 24 m. Calculate the length of the canvas used to make the tent. If the width of the rectangular canvas is 4 m.
41. A card is drawn from a pack of 52 cards. Find the probability of getting a king or a heart or a red card.
42. Find the coefficient of variation of the data 18, 20, 15, 12, 25.

PART - IV**Note : Answer the following questions..** **$2 \times 8 = 16$**

43.

- a) Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$.

(OR)

- b) The following table shows the data about the number of pipes and the time taken to fill the same tank.

No. of pipes (x)	2	3	6	9
Time Taken (in min) (y)	45	30	15	10

Draw the graph for the above data and hence

- (i) find the time taken to fill the tank when five pipes are used.
 (ii) find the number of pipes when the time is 9 minutes.

44.

- a) Draw $\triangle PQR$ such that $PQ = 6.8$ cm, vertical angle is 50° and the bisector of the vertical angle meets the base at D where $PD = 5.2$ cm.

(OR)

- b) Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR. (scale factor $\frac{3}{5} < 1$)



10TH MATHS PTA QUESTION MODEL -4

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

$$\boxed{14 \times 1 = 14}$$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

1. The range of $R = \{ (x, x^2) | x \text{ is a prime number less than } 13 \}$ is
 - 1) $\{4, 9, 25, 49, 121\}$
 - 2) $\{1, 4, 9, 25, 49, 121\}$
 - 3) $\{2, 3, 5, 7\}$
 - 4) $\{2, 3, 5, 7, 11\}$
2. Let $A = \{1, 2, 3, 4\}$, $B = \{4, 8, 9, 10\}$. A function $f: A \rightarrow B$ given by $f = \{(1, 4), (2, 8), (3, 9), (4, 10)\}$ is a
 - 1) Many-one function
 - 2) Identity function
 - 3) One to One function
 - 4) Into function
3. If 6 times of 6th term of an A.P., is equal to 7 times the 7th term, then the 13th term of the A.P. is
 - 1) 0
 - 2) 6
 - 3) 7
 - 4) 13
4. The sum of the exponents of prime factors in the prime factorization of 1729 is
 - 1) 4
 - 2) 3
 - 3) 2
 - 4) 1
5. If a and b are two positive integers where $a > 0$ and b is a factor of a , then HCF of a and b is
 - 1) b
 - 2) a
 - 3) $3ab$
 - 4) $\frac{a}{b}$
6. If $(x - 6)$ is the HCF of $x^2 - 2x - 24$ and $x^2 - kx - 6$ then the value of k is
 - 1) 8
 - 2) 6
 - 3) 5
 - 4) 3
7. If a polynomial is a perfect square then its factors will be repeated number of times.
 - 1) Odd
 - 2) Zero
 - 3) Even
 - 4) None of the above
8. If $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is
 - 1) $5\sqrt{2}$ cm
 - 2) 10 cm
 - 3) 2.5 cm
 - 4) 5 cm
9. When proving that a quadrilateral is a trapezium, it is necessary to show
 - 1) Two parallel and two non-parallel sides
 - 2) Two sides are parallel.
 - 3) Opposite sides are parallel
 - 4) All sides are of equal length
10. The equation of a line passing through the origin and perpendicular to the line $7x - 3y + 4 = 0$ is
 - 1) $7x - 3y + 4 = 0$
 - 2) $3x - 7y + 4 = 0$
 - 3) $7x - 3y = 0$
 - 4) $3x + 7y = 0$
11. If $\sin \theta = \cos \theta$ and $2\tan^2 \theta + \sin^2 \theta - 1$ is equal to
 - 1) $\frac{3}{2}$
 - 2) $-\frac{3}{2}$
 - 3) $\frac{2}{3}$
 - 4) $-\frac{2}{3}$
12. In a hollow cylinder, the sum of the external and internal radii is 14 cm and the width is 4 cm. If its height is 20 cm, the volume of the material in it is
 - 1) $56 \pi \text{ cm}^3$
 - 2) $3600 \pi \text{ cm}^3$
 - 3) $5600 \pi \text{ cm}^3$
 - 4) $11200 \pi \text{ cm}^3$
13. Which of the following is incorrect?
 - 1) $P(A) + P(\bar{A}) = 1$
 - 2) $P(\phi) = 0$
 - 3) $0 \leq P(A) \leq 1$
 - 4) $P(A) > 1$
14. Probability of getting 3 heads or 3 tails in tossing a coin 3 times is
 - 1) $\frac{1}{8}$
 - 2) $\frac{1}{4}$
 - 3) $\frac{3}{8}$
 - 4) $\frac{1}{2}$

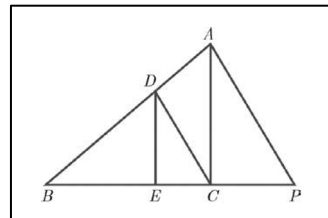
PART - II

Note : Answer **any 10** questions. Question **No. 28** is compulsory.

$$\boxed{10 \times 2 = 20}$$

- 15.** Find k if $f \circ f(k) = 5$ where $f(k) = 2k - 1$.
- 16.** Let $A = \{1, 2, 3, \dots, 100\}$ and R be the relation defined as “is cube of” on A . find the domain and range of R .

17. In a theatre there are 20 seats in the front row and 30 rows were allotted. Each successive row contains two additional seats than its front row. How many seats are there in the last row?
18. In a G. P $\frac{1}{4}, -\frac{1}{2}, 1, -2, \dots$ find t_{10} .
19. Which rational expression should be subtracted from $\frac{x^2+6x+8}{x^8+8}$ to get $\frac{3}{x^2-2x+4}$.
20. Determine the quadratic equation, whose sum and product of roots are $-\frac{3}{2}$ and -1 .
21. State Pythagoras theorem.
22. In the figure $DE \parallel AC$ and $DC \parallel AP$. Prove that $\frac{BE}{EC} = \frac{BC}{CP}$.
23. Show that the points $P(-1.5, 3), Q(6, -2), R(-3, 4)$ are collinear.
24. Prove that $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}$.
25. The volumes of two cones of same base radius are 3600 cm^3 and 5040 cm^3 . Find the ratio of their heights.
26. The range of a set of data is 13.67 and the largest value is 70.08. find the smallest value.
27. Write the sample space for selecting two balls from a bag containing 6 balls numbered 1 to 6 using tree diagram.
28. Find the sum and product of the roots of the equation $8x^2 - 25 = 0$.



PART - III

Note : Answer any 10 questions. Question No. 42 is compulsory.

10 × 5 = 50

29. The data in the adjacent table depicts the length of a woman's forehead and her corresponding height. Based on this data, a student finds a relationship between the height(y) and the forehead length (x) as $y = ax + b$, where a, b are constants
- (i) Check if this relation is a function (ii) Find a and b
- (iii) Find the height of a woman whose forehead length is 40 cm
- (iv) Find the length of forehead of a woman if her height is 53.3 inches.

Length x of forehead (in cm)	Height y (in inches)
35	56
45	65
50	69.5
55	74

30. A function $f: [-5, 9] \rightarrow \mathbb{R}$ is defined as follows $f(x) = \begin{cases} 6x + 1 & \text{if } -5 \leq x < 2 \\ 5x^2 - 1 & \text{if } 2 \leq x < 6 \\ 3x - 4 & \text{if } 6 \leq x \leq 9 \end{cases}$.
- Find (i) $f(7) - f(1)$ (ii) $\frac{2f(-2) - f(6)}{f(4) + f(-2)}$.

31. Find the sum to n terms of the series $5 + 55 + 555 + \dots$.
32. A girl is twice as old as her sister. Five years hence, the product of their ages (in years) will be 375. Find their present ages.
33. Find the non-zero values of x satisfying the matrix equation
$$x \begin{pmatrix} 2x & 2 \\ 3 & x \end{pmatrix} + 2 \begin{pmatrix} 8 & 5x \\ 4 & 4x \end{pmatrix} = 2 \begin{pmatrix} x^2 + 8 & 24 \\ 10 & 6x \end{pmatrix}.$$
34. Find the values of a and b if the polynomial $4x^4 - 12x^3 + 37x^2 + bx + a$ is a perfect square.
35. State and prove alternate segment theorem.
36. $PQRS$ is a rhombus. Its diagonals PR and QS intersect at the point M and satisfy $QS = 2PR$. If the coordinates of S and M are $(1,1)$ and $(2, -1)$ respectively, find the coordinates of P .

37. A building and a statue are in opposite side of a street from each other 35 m apart. From a point on the roof of building the angle of elevation of the top of statue is 24° and the angle of depression of base of the statue is 34° . Find the height of the statue. ($\tan 24^\circ = 0.4452$, $\tan 34^\circ = 0.6745$).
38. A cylindrical bucket, 32 cm high and with radius of base 18 cm, is filled with sand completely. This bucket is emptied on the ground and a conical heap of sand is formed. If the height of the conical heap is 24 cm, find the radius and slant height of the heap.
39. The consumption of number of guava and orange by a family on a particular week are given below. Which fruit is consistently consumed by the family?
- | | | | | | | | |
|-------------------|---|---|---|---|---|---|---|
| Number of Guavas | 3 | 5 | 6 | 4 | 3 | 5 | 4 |
| Number of Oranges | 1 | 3 | 7 | 9 | 2 | 6 | 2 |
40. In a class of 50 students, 28 opted for NCC, 28 opted for NSS and 10 opted for both NCC and NSS. One of the students is selected at random. Find the probability that
 (i) the student opted for NCC but not NSS. (ii) The student opted for NSS but not NCC.
 (iii) The student opted for exactly one of them.
41. By using slopes, show that the points $(1, -4)$, $(2, -3)$ and $(4, -7)$ form a right angled triangle.
42. A man saved ₹ 16,500 in ten years. In each year after the first he saved ₹ 100 more than he did in the preceding year. How much did he save the first year?.

PART - IV

Note : Answer the following questions..

$2 \times 8 = 16$

43.

- a) Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$.

(OR)

- b) Graph the following linear function $y = \frac{1}{2}x$. Identify the constant of variation and verify it with the graph. Also (i) find y when $x = 9$ (ii) find x when $y = 7.5$.

44.

- a) Draw a ΔPQR such that $PQ = 6.8$ cm, vertical angle is 50° and the bisector of the vertical angle meets the base at D , where $PD = 5.2$ cm.

(OR)

- b) Draw a ΔPQR in which $QR = 5$ cm, $\angle P = 40^\circ$ and the median PG from P to QR is 4.4 cm. Find the length of the altitude from P to QR .



10TH MATHS PTA QUESTION MODEL -5

Time Allowed: 3.00 Hours

Maximum Marks: 100

PART - I

Note : (i) Answer all the questions.

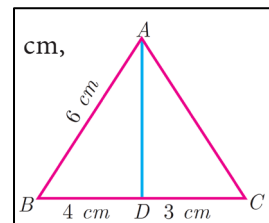
 $14 \times 1 = 14$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- $f(x) = (x + 1)^3 - (x - 1)^3$ represents a function which is
1) Quadratic 2) cubic 3) linear 4) reciprocal
- Using Euclid's division lemma, if the cube of any positive integer is divided by 9 then the possible remainders are
1) 1,3,5 2) 1,4,8 3) 0,1,3 4) 0,1,8
- An A.P. consists of 31 terms. if its 16th term is m , then the sum of all the terms of this A.P. is
1) $16m$ 2) $62m$ 3) $\frac{31}{2}m$ 4) $31m$
- $\frac{3y-3}{y} \div \frac{7y-7}{3y^2}$ is
1) $\frac{9y^3}{(21y-21)}$ 2) $\frac{9y}{7}$ 3) $\frac{21y^2-42y+21}{3y^3}$ 4) $\frac{7(y^2-2y+1)}{y^2}$
- The solution of $x^2 - 25 = 0$ is
1) No real roots 2) Real and equal roots
3) Real and unequal roots 4) Imaginary roots
- For the given matrix $A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$ the order of the matrix $(A)^{T^T}$ is
1) 2×3 2) 3×2 3) 3×4 4) 4×3
- The perimeters of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 36 cm and 24 cm respectively. If $PQ = 10$ cm, then length of AB is
1) $6\frac{2}{3}$ cm 2) $66\frac{2}{3}$ cm 3) $\frac{10\sqrt{6}}{3}$ cm 4) 15 cm
- If $(5, 7)$, $(3, p)$ and $(6, 6)$ are collinear, then the value of, p is
1) 9 2) 12 3) 3 4) 6
- If the points $A(6,1)$, $B(8,2)$, $C(9,4)$ and $D(p, 3)$ are the vertices of a parallelogram, taken in order then the value of p is
1) -7 2) 7 3) 6 4) -6
- $a \cot \theta + b \operatorname{cosec} \theta = p$ and $b \cot \theta + a \operatorname{cosec} \theta = q$ then, $p^2 - q^2$ is equal to
1) $a^2 + b^2$ 2) $a^2 - b^2$ 3) $b^2 - a^2$ 4) $b - a$
- The ratio of the volumes of a cylinder, a cone and a sphere, if each has the same diameter and same height is
1) 1:2:3 2) 3:1:2 3) 2:1:3 4) 1:3:2
- C.S.A of solid sphere is equal to
1) T.S.A of solid sphere 2) T.S.A of hemisphere
3) C.S.A of hemisphere 4) None of these
- Variance of first 20 natural numbers is
1) 32.25 2) 33.25 3) 44.25 4) 30
- Which of the following is incorrect?
1) $P(A) + P(\bar{A}) = 1$ 2) $P(\phi) = 0$ 3) $0 \leq P(A) \leq 1$ 4) $P(A) > 1$

PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

15. Let $A = \{1, 2, 3, 4\}$ and $B = \mathbb{N}$. Let $f: A \rightarrow B$ be defined by $f(x) = x^2$, find
 (i) The range of f (ii) identify the type of function.
16. If $3 + k, 18 - k, 5k + 1$ are in A.P then find k .
17. Find the geometric progression whose first term $a = -7$ and common ratio $r = 6$.
18. Find the square root of $\frac{144a^8b^{12}c^{16}}{81f^{12}g^4h^{14}}$.
19. Which term of the A.P 21, 18, 15, ... is -81 ? State with reason is three any term 0 in this A.P?.
20. A relation R is given by the set $\{(x, y)/y = x + 3, x \in \{0, 1, 2, 3, 4, 5\}\}$. Determine its domain and range.
21. If $A = \begin{pmatrix} 0 & 4 & 9 \\ 8 & 3 & 7 \end{pmatrix}$ and $B = \begin{pmatrix} 7 & 3 & 8 \\ 1 & 4 & 9 \end{pmatrix}$ find the value of $3A - 9B$.
22. In the adjacent figure AD is the bisector of A . If $BD = 4 \text{ cm}$, $DC = 3 \text{ cm}$ and $AB = 6 \text{ cm}$. Find AC .
23. Show that the straight lines $x - 2y + 3 = 0$ and $6x + 3y + 8 = 0$ are perpendicular.
24. Show that $\sqrt{\frac{\sec \theta - \tan \theta}{\sec \theta + \tan \theta}} = \frac{1 - \sin \theta}{\cos \theta}$.
25. If the radii of the circular ends of a frustrum which is 45 cm high are 28 cm and 7 cm, find the volume of the frustrum.
26. Find the range of the following distribution.
- | Age | 16 - 18 | 18 - 20 | 20 - 22 | 22 - 24 | 24 - 26 | 26 - 28 |
|--------------------|---------|---------|---------|---------|---------|---------|
| Number of students | 0 | 4 | 6 | 8 | 2 | 2 |
27. Three fair coins are tossed together. Find the probability of getting (i) Atleast one tail (ii) atmost one hand.
28. Find the value of p , when $px^2 + (\sqrt{3} - \sqrt{2})x - 1 = 0$ and $x = \frac{1}{\sqrt{3}}$ is one root of the equation.

**PART - III****Note : Answer any 10 questions. Question No. 42 is compulsory.****10 × 5 = 50**

29. Let $A = \{x \in W / x < 2\}$, $B = \{x \in N / 1 \leq x \leq 4\}$ and $C = \{3, 5\}$.
 Verify that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
30. If $f(x) = 2x + 3$, $g(x) = 1 - 2x$, $h(x) = 3x$, then prove that $f \circ (g \circ h) = (f \circ g) \circ h$.
31. A man repays a loan of ₹ 65, 000 by paying ₹ 400 in the first month and then increasing the payment by ₹ 300 every month. How long will it take for him to clear the loan?.
32. Find the sum of $10^3 + 11^3 + 12^3 + \dots + 20^3$.
33. Solve the system of linear equations in three variables: $x + y + z = 5$, $2x - y + z = 9$, $x - 2y + 3z = 16$.
34. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the value of a and b .
35. If $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$. Show that $A^2 - 4A + 5I_2 = 0$.
36. State and prove Angle Bisector Theorem.
37. Find the value of k , if the area of a quadrilateral is 28 sq. units, whose vertices are $(-4, -2)$, $(-3, k)$, $(3, -2)$ and $(2, 3)$.

38. Two ships are sailing in the sea on either sides of a lighthouse. The angle of elevation of the top of the lighthouse as observed from the ships are 30° and 45° respectively. If the lighthouse is 200 m high, find the distance between the two ships. $\sqrt{3} = 1.732$.
39. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice cream. The ice cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container.
40. A well of diameter 3 m is dug 14 m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4 m to form an embankment. Find the height of the embankment.
41. The time taken by 50 students to complete a 100 meter race are given below. Find its standard deviation.

Time taken (seconds)	8.5- 9.5	9.5-10.5	10.5-11.5	11.5-12.5	12.5-13.5
Number of Students	6	8	17	10	9

42. A card is drawn from a pack of 52 cards. Find the probability of getting a Queen or a diamond or a black card.

PART - IV

Note : Answer the following questions..

$2 \times 8 = 16$

43.

- a) A company initially started with 40 workers to complete the work by 150 days. Later, it decided to fasten up the work increasing the number of workers as shown below.

Number of workers (x)	40	50	60	75
Number of days (y)	150	120	100	80

- (i) Graph the above data and identify the type of variation.
- (ii) From the graph, find the number of days required to complete the work if the company decides to opt for 120 workers?
- (iii) If the work has to be completed by 30 days, how many workers are required?

(OR)

- b) Draw the graph of $y = x^2 + 3x - 4$ and hence use it solve $x^2 + 3x - 4 = 0$

44.

- a) Construct a triangle ΔPQR such that $QR = 5$ cm, $\angle P = 30^\circ$ and the altitude from P to QR is of length 4.2 cm.

(OR)

- b) Draw the two tangents from a point which is 10 cm away from the centre of a circle of radius 5 cm. Also, measure the lengths of the tangents.



10TH MATHS PTA QUESTION MODEL -6

Time Allowed: 3.00 Hours

Maximum Marks: 100

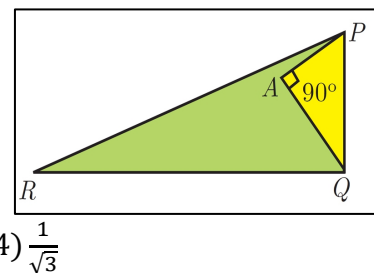
PART - I

Note : (i) Answer all the questions.

 $14 \times 1 = 14$

(ii) Choose the most appropriate answer from the given four alternative and write the option code and the corresponding answer.

- If $g = \{(1,1), (2,3), (3,5), (4,7)\}$ is a function given by $g(x) = \alpha x + \beta$; then the values of α and β
 - (1, 2)
 - (-1, 2)
 - (2, -1)
 - (-1, -2)
- The given diagram represents
 - an onto function
 - a constant function
 - an one-one function
 - not a function
- If $A = 2^{65}$ and $B = 2^{64} + 2^{63} + 2^{62} + \dots + 2^0$ which of the following is true?
 - B is 2^{64} more than A
 - B is larger than A by 1
 - A and B are equal
 - A is larger than B by 1
- If a, b, c are in A.P then $\frac{a-b}{b-c}$ is equal to
 - $\frac{a}{b}$
 - $\frac{b}{c}$
 - $\frac{a}{c}$
 - 1
- $y^2 + \frac{1}{y^2}$ is not equal to
 - $[y - \frac{1}{y}]^2 + 2$
 - $[y + \frac{1}{y}]^2 - 2$
 - $[y + \frac{1}{y}]^2$
 - $\frac{y^4+1}{y^2}$
- Find the matrix X if $2X + \begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 5 & 7 \\ 9 & 5 \end{pmatrix}$
 - $\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$
 - $\begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}$
 - $\begin{pmatrix} -2 & -2 \\ 2 & -1 \end{pmatrix}$
 - $\begin{pmatrix} 2 & 2 \\ 2 & -1 \end{pmatrix}$
- On dividing $\frac{x^2-25}{x+3}$ by $\frac{x+5}{x^2-9}$ is equal to
 - $(x-5)(x-3)$
 - $(x-5)(x+3)$
 - $(x+5)(x-3)$
 - $(x+5)(x+3)$
- In $\triangle ABC$, AD is the bisector of $\angle BAC$. If $AB = 8\text{ cm}$, $BD = 6\text{ cm}$ and $DC = 3\text{ cm}$. The length of the side AC is
 - 3 cm
 - 4 cm
 - 6 cm
 - 8 cm
- In a given figure, $PR = 26\text{ cm}$, $QR = 24\text{ cm}$, $\angle PAQ = 90^\circ$, $PA = 6\text{ cm}$, $QA = 8\text{ cm}$. Find $\angle PQR$
 - 90°
 - 85°
 - 80°
 - 75°
- If slope of the line PQ is $\frac{1}{\sqrt{3}}$, then slope of the perpendicular bisector of PQ is
 - 0
 - $\sqrt{3}$
 - $-\sqrt{3}$
 - $\frac{1}{\sqrt{3}}$
- If the ratio of the height of a tower and the length of its shadow is $\sqrt{3} : 1$, then the angle of elevation of the sun has measure
 - 90°
 - 60°
 - 45°
 - 30°
- A spherical ball of radius r_1 units is melted to mark 8 new identical balls each of radius r_2 units. Then $r_1 : r_2$ is
 - 1: 4
 - 4: 1
 - 1: 2
 - 2: 1
- A fair die is thrown once. The probability of getting a prime (or) composite number is
 - 1
 - 0
 - $\frac{5}{6}$
 - $\frac{1}{6}$
- Which of the following is not a measure of dispersion?
 - Range
 - Standard deviation
 - Arithmetic mean
 - Variance



PART - II**Note : Answer any 10 questions. Question No. 28 is compulsory.****10 × 2 = 20**

15. Let f be a function from $\mathbb{R}$ defined by $f(x) = 3x - 5$. Find the values of a and b given that $(a, 4)$ and $(1, b)$ belong to f .
16. If $R = \{(x, -2), (-5, y)\}$ represents the identify function, find the values of x and y .
17. Find the common difference of an A.P in which $t_{18} - t_{14} = 32$.
18. Find the number of integer solutions of $3x \equiv 1 \pmod{15}$.
19. Find the sum of $1 + 3 + 5 + \dots + 55$.
20. Solve by factorization method : $2x^2 - 2\sqrt{6}x + 3 = 0$.
21. If the difference between a number and its reciprocal is $\frac{24}{5}$, find the number.
22. If α, β are the roots of the equation $7x^2 + ax + 2 = 0$ and if $\beta - \alpha = -\frac{13}{7}$, then find the value of a .
23. The line through the points $(-2, 6)$ and $(4, 8)$ is perpendicular to the line through the points $(8, 12)$ and $(x, 24)$. Find the value of x .
24. From the top of a rock $50\sqrt{3}$ m high, the angle of depression of a car the ground is observed to be 30° . Find the distance of the car from the rock.
25. A solid sphere and a solid hemisphere have equal total surface area. Prove that the ratio of their volume is $3\sqrt{3} : 4$.
26. Find the standard deviation of first 21 natural numbers.
27. A and B are two candidates seeking admission to IIT. The probability that A getting selected is 0.5 and the probability that both A and B getting selected is 0.3. prove that the probability of B being selected is at most 0.8.
28. P and Q are points on sides AB and AC respectively of $\triangle ABC$. If $AP = 3$ cm, $PB = 6$ cm, $AQ = 5$ cm and $QC = 10$ cm. Show that $BC = 3 PQ$.

PART - III**Note : Answer any 10 questions. Question No. 42 is compulsory.****10 × 5 = 50**

29. Write the domain of the following functions: (i) $\frac{2x+1}{x-9}$ (ii) $g(x) = \sqrt{x-2}$.
30. If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ are defined by $f(x) = x^5$ and $g(x) = x^4$ then check if f and g are one - one and $f \circ g$ is one - one?.
31. If the sum of the first p terms of an A.P is $ap^2 + bp$. Find its common difference.
32. A man joined a company as Assistant Manager. The company gave him a starting salary of ₹ 60,000 and agreed to increase his salary 5% annually. What will be his salary after 5 years?.
33. If the roots of the equation $(c^2 - ab)x^2 - 2(a^2 - bc)x + b^2 - ac = 0$ are real and equal, prove that either $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.
34. Find the LCM of the polynomials $a^2 + 4a - 12$, $a^2 - 5a + 6$ whose GCD is $a - 2$.
35. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 3 \\ -1 & 5 \end{pmatrix}$, $C = \begin{pmatrix} -1 & 5 \\ 1 & 3 \end{pmatrix}$. Prove that $A(BC) = (AB)C$.
36. The perpendicular PS on the base QR of a $\triangle PQR$ intersects QR at S , such that $QS = 3SR$. Prove that $2PQ^2 = 2PQ^2 + QR^2$.
37. Find the equation of the median and altitude of $\triangle ABC$ through A where the vertices are $A(6, 2)$, $B(-5, -1)$ and $C(1, 9)$.

38. Prove that $\left[\frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} \right] - \left[\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} \right] = 2 \sin A \cos A$.
39. If the slant height of the frustum cone is 10 cm and perimeters of its circular base are 18 cm and 28 cm respectively. What is the curved surface area of the frustum?.
40. A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice cream. The ice cream is to be filled in cones of height 9 cm base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container.
41. The following table gives the values of mean and variance of heights and weights of the 10th standard students of a school. Which is more varying than the other.

	Height	Weight
Mean	155 cm	46.50 kg
Variance	72.25 cm ²	28.09 kg ²

42. A coin is tossed thrice. Find the probability of getting exactly two heads or at least one tail or two consecutive heads.

PART - IV

Note : Answer the following questions..

2 × 8 = 16

- 43.
- a) Draw the two tangents from a point which is 10cm away from the centre of a circle of radius 5 cm. Also, measure the lengths of the tangents.

(OR)

- b) Construct a triangle similar to a given triangle PQR with its sides equal to $\frac{3}{5}$ of the corresponding sides of the triangle PQR (scale factor $\frac{3}{5} < 1$)

44.

- a) Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$

(OR)

- b) The following table shows the data about the number of pipes and the time taken to fill the same tank.

No. of pipes (x)	2	3	6	9
Time Taken (in min) (y)	45	30	15	10

Draw the graph for the above data and hence

- (i) find the time taken to fill the tank when five pipes are used
(ii) Find the number of pipes when the time is 9 minutes.



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